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PAY THE DEVIL IN BITCOIN

A Kindle Single

PAY THE DEVIL IN BITCOIN

The Creation of a Cryptocurrency and How Half a Billion Dollars of It Vanished from Japan

JAKE ADELSTEIN AND NATHALIE STUCKY

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PREFACE

BY JAKE ADELSTEIN

There's a saying in Japanese—*Jigoku no sata mo kane shidai*—meaning "Even at the gates of hell, what happens depends on how much money you have." The saying is the inspiration for the title of this book and illustrates something we all know is true: we are treated very differently according to how much wealth we have. It can be the difference between heaven or hell.

Some may ask, does hell really exist? Isn't that a matter of metaphysical belief?

Yes, it is. And in some ways, belief in heaven or hell is about as rational as belief in a cryptocurrency called bitcoin.

Bitcoin is not just a cryptocurrency; it is a financial force, and the faith placed in its power and probity by true believers is what gives it value. In that sense, it is almost magical.

There have been several excellent books written about bitcoin and a few good documentaries. This book is for people who are curious about this cryptocurrency and want to know more but don't need a 350-page tome on the subject.

It's also a book about bitcoin and Japan. The two are interlinked in many ways: the mysterious founder has a Japanese name, and the first major bitcoin exchange, Mt. Gox, was established in Tokyo. Japan is also home to one of bitcoin's foremost missionaries, Roger Ver, jokingly referred to as the "Bitcoin Jesus."

The Swiss journalist Nathalie Stucky and I have spent two years covering the rise and fall and rise again of bitcoin and the collapse of what was once the center of the bitcoin universe, Mt. Gox. Most of our articles covering the story were originally published in the *Daily Beast*, an American website focusing on politics, world affairs, and pop culture, which currently reaches more than twenty million readers per month. As time has gone by, we have become close to many of the people appearing in this book. Hopefully, this hasn't affected our judgment. Ms. Stucky in particular became very close to and very fond of Mark Karpelès, the de facto creator of Mt. Gox, and the fondness appeared to be mutual. I'm about as fond of Karpelès as I am of his cats, but I do believe that he has been dealt an unfair hand.

The gods of fortune in Japan are very fickle, although apparently they will take bribes. Even if you bribe them with money or commodities,

however, they don’t always deliver.

One of the first things you’re told as a journalist about the basics of investigative journalism is this: follow the money. If you follow the bitcoin, you are likely to encounter a strange alternate world populated by shady characters, drug dealers, diligent special agents, corrupt cops, hapless coders, and pirate hackers. It’s a fun world to explore.

Bitcoin doesn’t exist in solid form like most fiat money. It’s shapeless, immaterial, usually traded over the Internet. Yet it wasn’t created out of nothing—the currency rests on complicated computer programming, a public ledger, a dedicated group of users—and it is given value by scarcity, trust, and above all human greed.

And as long as the human race is greedy, I don’t think we’ll see bitcoin vanish from our world. The currency may be new, but the raw desire behind it has been around for ages.

You can pay the devil in bitcoin. And since 2013, at least one church in New York City accepts donations in bitcoin as well.

A NOTE ON SOURCES

This book was written with the cooperation of former Mt. Gox employees, including CEO Mark Karpelès. Its creation involved extensive interviews with lawyers, journalists, police officers, and even people inside the prosecutor's office, who went above and beyond the call of duty and sometimes against their own self-interests to share with us details of what they knew. We have added citations for those who would like to know more, but we have not given dates for each and every interview we conducted. Suffice it to say that when quotations are given without sources, they come from real conversations we had with our informants.

The people we spoke with told us, in great detail, about the Mt. Gox case and related criminal cases, some of which are still ongoing. Some sources provided documents. In doing so, they took incredible risks. Some of them violated NDAs or rules set up by their organizations that forbade them from speaking to the press without permission, and, possibly, in speaking to us they broke the law. If they were caught cooperating with us or if the full contents of what they disclosed were to become public, they could face not only the loss of their jobs but prosecution for violations of Japan's civil-servants law (for disclosure of information gained in capacity as a public official) or even retaliation from the firms that employ them.

Therefore, in order to protect sources, we have changed some names and obscured certain details. Bitcoin is a virtual currency, but our sources are real people. We would like to ensure their well-being, at least, and to thank here those who can't be named—thank you!

CHAPTER ONE

BITCOIN AND ITS MYSTERIOUS FOUNDER

We use money every day, and we never stop to think about it. While we may be moving toward a cashless society, we still carry around wallets containing one-dollar bills with the face of George Washington on them, or hundred-yen coins engraved with cherry blossoms, and we expect the value of what we're carrying to remain fairly constant. Currency traders make money on guessing when my hundred-yen coin is going to be worth less or more than a dollar bill.

At this point in time, the rate is about one hundred yen to the dollar. It makes calculating things easy. But that won't last for long.

The US dollar is still the gold standard of world currency, but the value of a currency can fluctuate tremendously, and it often depends on the way governments manipulate the market or on external political factors that can't be controlled.

When the result of the Brexit vote was Britain deciding to leave the EU, the euro and the pound plummeted. That was not a planned move.

In Japan, Prime Minister Shinzo Abe's economic plan, dubbed "Abenomics," drove down the value of the yen for an extended period of time.

For centuries, almost every currency in the world has been issued by governments, which have considerable power to affect its value and even regulate its supply.

And then along came bitcoin (BTC) in 2008. It's not backed by any government. There is no Royal Bank of Bitcoin. Yet increasingly it has been accepted as a valid form of payment.

What is bitcoin? A dummy currency printed on poker chips? A Ponzi scheme? The wet dream of criminal enterprises?

No. Bitcoin is the name of a digital cryptocurrency created by a volunteer network of computer users. It is held electronically. It aims to be a secure method of buying items and sending money without the transaction fees of services like PayPal. If you don't want to spend forty dollars transferring a small amount of money out of Japan to the United States, use bitcoin instead. To buy it or sell it, all you need to do is set up an online account. On coinbase.com, this can be done in a matter of minutes. You can use other online exchanges to buy it with your credit card, or you can buy it

from someone with hard cash. You can even use free bitcoin software to set up your own virtual wallet to receive money by logging into blockchain.info.

The appeal of bitcoin is that it liberates you from the conventional banking system. And you don't actually need to understand how the digital currency works in order to use it, just as you don't need to be an expert to use a computer.

Bitcoin may have started as an experiment, but it has become a genuine player in the financial world and the world at large, even gaining recognition as a word by the stodgiest of sources. Along with the term "twerk," it was listed in the Oxford Dictionaries in its quarterly update in August 2013. This delighted members of the bitcoin community, although some expressed disappointment that the word *satoshi*—the smallest denomination of bitcoin possible—didn't make the cut. The dictionary defines the word as "[a] type of digital currency in which encryption techniques are used to regulate the generation of units of currency and verify the transfer of funds, operating independently of a central bank."

Angus Stevenson, head of dictionary projects there, has said that the criterion for what word makes it into the publication is strict. "Inclusion in the ODO provides a recognition that a term has entered public consciousness and wide public use, it doesn't make any judgement on whether it is good, bad, worthwhile or anything else.

"It's just that we feel that, as a word, it is in genuine wide use and we feel it will stick around as part of the language. It may become a byword for a failed scheme, but we don't know in real terms what will become of it."[1]

It has indeed seemed possible that it might end up as "a byword for a failed scheme."

Bitcoin attracted great media attention after the collapse, in February 2014, of the world's first large-scale bitcoin exchange, Mt. Gox. Mt. Gox entered bankruptcy protection on February 28, 2014, after asserting that it had been hacked and had lost nearly 850,000 BTC with an estimated value of over $460 million at the time. The actual amount that was lost, hacked, or stolen eventually turned out to be 650,000 BTC, after 200,000 BTC was later recovered (still an astronomical amount of money). Hundreds of thousands of people lost money in the collapse. Some individuals lost $150,000; some claim to have lost millions. A month after the collapse, the Japanese police opened a formal investigation into what happened to the missing amount at the request of Mark Karpelès, who provided them with all the data he had.

Only 1 percent of those who lost money were Japanese.

This is ironic because Japan is essentially the birthplace of bitcoin. Its enigmatic creator, Satoshi Nakamoto, is supposedly Japanese. Mt. Gox, which transformed the currency from a novelty into a money market staple, was located in Tokyo.

The Japanese government now regulates the currency and the transactions related to it. Officials are also apparently looking for Nakamoto-san, whose real identity is unknown. Why? Perhaps to give him an award or, alternatively, arrest him.

It's hard to talk about bitcoin without Nakamoto coming to mind. It's like trying to talk about Christianity without bringing Jesus into the picture.

We are assuming here that this person is a man, but he could be a "she" or even a "they." One could say he/she/they is the most sought-after figure in the information technology world.

Bitcoin was essentially born on October 31, 2008, when Nakamoto published a scientific paper titled "Bitcoin: A Peer-to-Peer Electronic Cash System" for a cryptography mailing list, describing the currency and the way the problem of double spending can be resolved to prevent the currency from being copied. Cryptography is the art of creating secret and secure means of communication, and it fascinates many programmers. The paper proposed a digital currency that could be "sent directly from one party to another without going through a financial institution."

However, while bitcoin was heralded as something that might liberate the financial markets in the way the Internet liberated information, there were some inherent problems. Ironically, it was Special Agent Tigran Gambaryan of the Internal Revenue Service (IRS) who in a March 2015 affidavit gave the best definition of bitcoin available for the layman, a definition made so simple that even a federal judge could understand. This isn't to imply that federal judges are stupid; they're just not computer science experts.[2] Read closely and you'll understand more about bitcoin than many journalists writing about it, and perhaps many people who trade it.

> Bitcoin is a form of decentralized, convertible virtual currency that exists through the use of an online, decentralized ledger system [the blockchain]. While Bitcoin mainly exists as an internet-based form of currency, it is possible to "print out" the necessary information and exchange Bitcoin via a physical medium [cold storage, paper wallets]. The currency is not issued by any government, bank, or company, but rather is generated and controlled through computer software operating via a decentralized network. To acquire bitcoins, a typical user will purchase them

from a Bitcoin seller or “exchanger.” It is also possible to “mine” bitcoin by verifying other users’ transactions. Bitcoin is just one form of digital currency, and there are a significant number of other varieties of digital currency.

Bitcoin exchangers typically accept payments of fiat currency (currency which derives its value from government regulation or law), or other convertible virtual currencies in order to obtain bitcoins. When a user wishes to purchase bitcoins from an exchanger, the user will typically send payment in the form of fiat or other convertible virtual currency to an exchanger, usually via wire or ACH, for the corresponding number of bitcoins based on a fluctuating exchange rate. The exchanger, often for a commission, will then typically attempt to broker the purchase with another user of the exchange that is trying to sell bitcoins, or, in some instances, will act as the seller itself. If the exchanger can place a buyer with a seller, then the transaction can be completed.

When a user acquires bitcoins, they are sent to the user’s Bitcoin address. This is somewhat analogous to a bank account number [or a safety-deposit box], which is comprised of a case-sensitive string of letters and numbers amounting to a total of 26 to 35 characters. The user can then conduct transactions with other Bitcoin users, by transferring bitcoins to their Bitcoin addresses, via the internet.

Little to no personally identifiable information about the payer or payee is transmitted in a Bitcoin transaction. Bitcoin transactions occur using a public key and a private key. A public key is used to receive bitcoins and a private key is used to allow withdrawals from a Bitcoin address. Only the Bitcoin address of the receiving party and the sender’s private key are needed to complete the transaction, which by themselves rarely reflect any identifying information.

All Bitcoin transactions are recorded on what is known as the block chain. This is essentially a distributed public ledger that keeps track of all Bitcoin transactions, incoming and outgoing, and updates approximately six times per hour. The block chain records every Bitcoin address that has ever received a bitcoin and maintains records of every transaction and all the known balances for each Bitcoin address.

Digital currencies, including Bitcoin, have many known legitimate uses. However, much like cash, bitcoins can be used to

> facilitate illicit transactions and to launder criminal proceeds, given the ease with which they can be used to move money anonymously. As is demonstrated herein, however, in some circumstances bitcoin payments may be traced to accounts at traditional financial institutions using the block chain.[3]

The idea of public keys and private keys is a little hard to grasp. Think of it this way: the public key is like a safety-deposit box that's open to the public. Anyone can put money into the box, but only the person with the private key (combination) can take the money out. If you want to take out the bitcoin deposited, you'll of course need to know where the safety-deposit box is and also have the combination. If you don't have both, your bitcoin is just about worthless.

A digital currency seemed like a good idea to many programmers, but there was always a major issue: digital items are easy to reproduce, easier than coinage or paper money. The act of spending digital money presents a problem if it does not remove the data from its owner. It's like when you send a digital photo from your cell phone to a friend via e-mail, you duplicate the digital data: there is one version of your photo on your phone and the same data on your friend's. Nakamoto came up with a brilliant solution in early 2009 by implementing the scheme called "proof-of-work" that would require users to verify all transactions in order to generate the currency and have it all recorded on a public ledger, so the users could police the system themselves and be rewarded for doing so.

Bobby Lee, the chief executive of China's best-known bitcoin exchange, BTCC (formerly BTCChina), explained it this way when we interviewed him:

> What happened with bitcoin that is revolutionary in the digital world, is that it allowed [you] to send something via the internet, like you send an email, except that when you send an email both the recipient and the expeditor have the email; but with bitcoin, when you send ten coins to someone, there are ten less in your wallet, and ten additional coins in the [other] person's wallet. And that is a powerful difference. It was impossible before bitcoin came along. For example, on Microsoft Word, it is the basic difference between "copy/paste" and "cut/paste."

"Bitcoin," Lee said, "is truly the world's first global currency that is for *the people* . . . It transcends geography, language, culture . . . any person, male or female, young or old. All you have to do is to have the internet . . . to hold bitcoins, send or receive bitcoins."

In the 2008 paper, Nakamoto wrote the following explanation. It's dense

and technical but outlines the core components of bitcoin.

> [Internet commerce reliant on financial institutions] works well enough for most transactions, [but] it still suffers from the inherent weaknesses of the trust based model. Completely non-reversible transactions are not really possible, since financial institutions cannot avoid mediating disputes. The cost of mediation increases transaction costs, limiting the minimum practical transaction size and cutting off the possibility of small casual transactions, and there is a broader cost in the loss of ability to make non-reversible payments for non-reversible services. With the possibility of reversal, the need for trust spreads. Merchants must be wary of their customers, hassling them for more information than they would otherwise need. A certain percentage of fraud is accepted as unavoidable. These costs and payment uncertainties can be avoided in person by using physical currency, but no mechanism exists to make payments over a communications channel without a trusted party.
>
> [Bitcoin] is an electronic payment system based on cryptographic proof instead of trust, allowing any two willing parties to transact directly with each other without the need for a trusted third party. Transactions that are computationally impractical to reverse would protect sellers from fraud, and routine escrow mechanisms could easily be implemented to protect buyers. In this paper, we propose a solution to the double-spending problem using a peer-to-peer distributed timestamp server to generate computational proof of the chronological order of transactions.[4]

One of Nakamoto's most interesting claims was that the currency did not require a third-party manager. Bitcoin, as explained in his paper, relies on a community of users to verify all transactions and generate the coins by using a shared computer network. So, instead of a bank, the currency uses a peer-to-peer (P2P) system. This means the work is multiplied by all the members of the network who perform the necessary tasks that a bank usually does.

Bitcoin uses its massive P2P network to verify transactions with the blockchain. The files within the chain, which detail every transaction, are called "blocks." The blockchain does not contain the names of buyers and sellers, only their bitcoin addresses, which are random sets of numbers and letters. One individual will usually "own" many bitcoin addresses. This helps keep bitcoin users' transactions private. In theory, and let's emphasize *in theory*, it also helps bitcoin users buy or sell whatever they want without fear

of being identified.[5] That's why it became the currency of choice for people buying child pornography, drugs, weapons, and other illicit items and services online. This isn't something covered in Nakamoto's paper.

There are rumors that Nakamoto currently owns as much as 1.5 million BTC. That's a gargantuan amount, considering the system was designed so that no more than 21 million BTC could ever be generated. Bitcoins are created, or "mined," by verifying transactions. In this way, it helps to think of bitcoin mining as a way for bank tellers to get a percentage of the take for the number of transactions they successfully complete and a sort of bonus for making sure the bank is not defrauded. Except that anyone, not just tellers, can mine.

The whole idea of mining might seem abstruse. Think of it as a way of stimulating a very high level of computer networking. Mining needs to be difficult in order to make the blockchain secure. No single individual should be strong enough to mine many blocks in a row. Because of the significant cost, an equivalent reward is set up, which also ensures a fair distribution of coins to those really involved in bitcoin. In its early days, mining spurred a speculative trade and competition to make a lot of money, especially as the price of bitcoin rose.[6]

Nakamoto not only came up with the idea for bitcoin but also was involved in developing the currency during its first few years. He was active on public bitcoin forums and corresponded with a number of developers who were dedicated to improving the software.

One such programmer was Gavin Andresen, who is now the chief scientist at the Bitcoin Foundation, a nonprofit corporation established in 2012 to standardize and advocate the use of the digital currency. Andresen was passionate about bitcoin and created the bitcoin faucet: a site that gave five free bitcoins to anyone who registered, in the belief that distributing the currency would help popularize it. Early on, he e-mailed Nakamoto to recommend a number of improvements to the system. Nakamoto trusted him, and they began to work together on the open-source program.

It took a number of years for bitcoin to go from theory to reality. The first real-world transaction occurred in May 2010, when Laszlo Hanyecz, a programmer living in Florida, bought two pizzas for 10,000 BTC. Hanyecz sent the coins to a fellow enthusiast in England, who ordered the pizzas for him from Papa John's using a credit card. In hindsight, these may have been the most expensive pizzas in the world.

The bitcoin market really began in July 2011, after Mark Karpelès, a French expat living in Tokyo, took what was originally an online exchange

for the card game Magic: The Gathering and turned it into what would become, for a few years, the largest and best-known bitcoin exchange in the world.

Bitcoins began to trade publicly with a thousand coins having a value of less than a dollar. By November, the bitcoin market value had passed $1 million. In December, at Nakamoto's request, Andresen's e-mail was posted on the official website, bitcoin.org, which was created on August 18, 2008.[7] Nakamoto was a prolific writer for a period of about two years, and *The Book of Satoshi: The Collected Writings of Bitcoin Creator Satoshi Nakamoto* includes a collection of his papers and e-mails for those who want the bitcoin equivalent of the New Testament's Q source.

However, Nakamoto was becoming markedly less active online by the end of 2010. His last e-mail before fading into the shadows for a few years was in 2011, and it was sent to a software developer. On the subject of bitcoin, he said, "I've moved on to other things. It's in good hands with Gavin [Andresen] and everyone."

It didn't matter who he was at that point or whether he was no longer actively involved. Bitcoin, like Frankenstein's monster, had taken on a life of its own.[8] On January 11, 2011, one bitcoin was worth thirty cents on the Mt. Gox exchange.

Technology reporters started to pay attention to the cryptocurrency as well. On February 10 the same year, a forum post on the tech website Slashdot had the headline "Online-Only Currency BitCoin Reaches Dollar Parity."[9]

Yes, one bitcoin was now worth a dollar.

Laszlo Hanyecz's pizzas were now worth $10,000—although presumably he'd eaten them by then. A Slashdot forum user known as "IamTheRealMike" wrote in the forum:

> The BitCoin peer to peer currency briefly reached exchange parity with the US dollar today after a spike in demand for the coins pushed prices slightly above 1 USD:1 BTC. BitCoin was launched in early 2009, so in only two years this open source currency has gone from having no value at all to one with not only an open market of competing exchanges, but the ability to buy real goods and services like web hosting, gadgets, organic beauty products and even alpaca socks.[10]

Things were heating up. April 2011, *Forbes* profiled bitcoin in an article titled "Crypto Currency."[11] That June, WikiLeaks, which had been banished from

the financial world by US banks, began accepting bitcoins as donations. By March 2013, the bitcoin market cap was over $1 billion. The value of bitcoin surged and surged as the financial crisis in Cyprus made people wary of banks. On April 1, 2013, bitcoin was worth a hundred dollars a coin. On April 9, the value doubled.

Despite mounting problems, security issues, and troubles galore at Mt. Gox—by this time the principal exchange—bitcoin continued to rise in price. It passed $1,000 and briefly held the same value as gold, trading at $1,240 per coin, thus earning the currency the moniker of “digital gold.” Bitcoin was big business. Governments were trying to decide how to regulate it. Bitcoin ATMs were appearing. Nakamoto’s invention was on the verge of being the next big thing.

However, the party wouldn’t last forever. After weeks of speculation that the exchange was in trouble, when Mt. Gox went blank on February 25, 2014, and filed for bankruptcy protection three days later, followed by a mythic press conference, the alchemy of fear turned this digital gold into virtual lead.

The perceived collapse of bitcoin that resulted, and the shocking realization that its principal exchange had lost nearly half a billion dollars, now made everyone curious about the currency and its creator.

A few weeks after Mt. Gox collapsed, *Newsweek* published a cover story titled “The Face behind Bitcoin” that claimed they had tracked down the legendary founder and confronted him at his modest home in a suburb of Los Angeles.[12] When the journalist behind the story, Leah McGrath Goodman, spoke in the driveway with somebody named Dorian Satoshi Nakamoto, the man told her he was “no longer involved in that and [couldn’t] discuss it.” And maybe that was enough proof for her.

After the story was published, the media swarmed down to Dorian’s place, but the sixty-four-year-old denied his involvement. He had simply been misunderstood by the reporter and didn’t even have a working Internet connection in the house.

On March 9, he gave an exclusive interview to the *Yomiuri Shimbun*, Japan’s largest newspaper. In it, he flatly denied being the creator of bitcoin; he said, “I had never even heard of bitcoin until a few weeks ago.”[13]

Confronted by the American reporter, he had thought she was asking him about classified work he had previously done for the US military, which he wasn’t entitled to discuss.

There were in fact many inconsistencies in the theory that Dorian was

Nakamoto, though *Newsweek* seemed to overlook them. For instance, bitcoin analysts are certain the real Nakamoto had mined a fortune for himself of up to a million bitcoin or more; in today's currency that would be valued at upward of a billion dollars. After the *Newsweek* story was published, new details emerged about Dorian, including that the bank had foreclosed his home and, due to various medical problems, he was openly in need of money.

Early on, the bitcoin community had speculated that Nakamoto was not actually Japanese, due to the fact that he never used the Japanese language. As it happens, he wrote well in English and used British spelling. Dorian's strongest language by far was his first: Japanese.

Dorian also mentioned something that a lot of other people were thinking: "The designer of bitcoin would never use his real name." After all, Nakamoto had gone to some lengths to anonymize all his personal information online.

The name itself, "Satoshi Nakamoto," is problematic. In the original 2008 paper that launched the virtual currency, there were no kanji representing Nakamoto's name. Kanji is a system of Japanese writing that uses Chinese characters to phonetically represent words, including names. Because of the limited number of vowels in Japanese, the language is riddled with homonyms. It is often extremely difficult to distinguish individuals or subjects without knowing the kanji characters. For example, the word *hashi* in Japanese could be represented as 箸 (chopsticks), 橋 (bridge), 端 (the tip), 嘴 (the beak of a bird), or 波子 (a train station in Shimane Prefecture).

In identifying an individual, the correct kanji are essential. This is why when the FBI asks Japan's National Police Agency for the names of yakuza gang members (Japan's mafia), they always request the kanji version of the person's name and their date of birth to avoid any mistake.

The first appearance of the kanji for Satoshi Nakamoto was in 2011, in a translation of the original essay posted on the bitcoin.co.jp website, which is part of the loosely connected federation of bitcoin enthusiasts. The kanji characters shown there are 中本 (Nakamoto) 哲史 (Satoshi). In traditional Japanese style, the family name is listed first.

There are hundreds of ways of representing Nakamoto Satoshi in kanji. Experts calculate that there are over thirty-eight thousand Nakamotos (中本) in Japan. However, when we were looking at the mystery of Nakamoto in March 2014, after the *Newsweek* article, in a twenty-year database search of over one hundred and fifty local newspapers and magazines in which the name 中本哲 appeared, there were only fourteen hits and only two individuals with that exact name (unrelated to articles about bitcoin). One was

a thirteen-year-old middle school student. The other was a teacher and baseball coach in his forties.

We contacted all those associated with bitcoin.co.jp to figure out why someone chose to use those specific kanji. The translation of the essay was finished on October 11, 2011, at 6:03 p.m., according to metadata embedded in the file. It isn't clear whether Nakamoto was consulted about the correct kanji use for the phonetic notation of his name. It is also not clear whether Momoko X, the translator of the essay, simply selected the kanji on her own. It is not even clear that the translator was not Satoshi. It appears that the characters were just randomly created. Mark Karpelès, the former CEO of Mt. Gox, claims to have asked Momoko, his employee at the time, to translate the paper.

This is probably true.

One thing you learn if you've written for Japanese publications for more than twenty years is that the standard practice for representing Japanese-named individuals when the kanji are unknown is to use katakana. Katakana is a simpler set of characters used mostly for nonnative Japanese words. Indeed, in the *Yomiuri Shimbun* article, the kanji for Dorian Satoshi Nakamoto don't actually appear. Instead, he is introduced in katakana as ドリアン・サトシ・ナカモト.

On March 7, 2014, the real Nakamoto posted the following through his long-dormant account at the P2P Foundation (the foundation for peer-to-peer alternatives): "I am not Dorian Nakamoto."

Of course, this could possibly have been done by Dorian himself, if he is the cyber genius behind bitcoin.

Prior to this message, Nakamoto's last post on the forum had been a reply to another member on February 18, 2009. Then, for five years, he remained silent.

The return of Nakamoto to the P2P Foundation forum raised a new mystery. Nakamoto first publicly posted the bitcoin software and an accompanying essay on February 11, 2009, at 10:27 p.m. There is a link to the essay, but when you download this seminal paper, stored as a pdf, the metadata indicates it was created on March 25, 2009, at 2:33:15 a.m.

Why does the paper posted there now have a misleading date? Did the original paper contain something compromising or incorrect and it was switched for a newer version?

Where is the original essay, if it exists? Was there something in the metadata that revealed Nakamoto's real identity?

Rashomon, anyone?[14] The classic Kurosawa film has become a cliché for a multiply confusing event. But in this case it actually sums up this tangled narrative perfectly. The film presents several contradictory accounts of the murder of a samurai in a forest thicket. The accounts include those of a prisoner, a woodcutter, a mendicant Buddhist priest, and even the ghost of the dead man himself, who questions whether the objective truth is to be found at all. Just so with Satoshi Nakamoto, whose identity seems increasingly lost in a cyber thicket that no one can penetrate.

CHAPTER TWO

MEET MAGICALTUX

Satoshi Nakamoto may have been the creator of bitcoin, but for several years the most important person in the bitcoin world was a Frenchman living in Japan.

The sudden shutdown of Mt. Gox in February 2014 drew a lot of attention to its CEO, Mark Karpelès, a.k.a. “MagicalTux.” His online identity was inspired by the Linux penguin mascot named Tux, and Karpelès used it on Twitter and in various other chat forums from an early stage.

So, who is he really?

Even those close to him don’t quite seem to get him.

“Mark Karpelès is like a more stoic version of the Cheshire Cat. There was always a strange disconnect when speaking with him. It made me wonder how this guy could be the Emperor of the Bitcoin Universe for even a day,” says a former accountant for Mt. Gox.

Born on June 1, 1985, Mark Marie Robert Karpelès is an entrepreneur; possibly a criminal mastermind (unlikely); depending on your point of view, either a shameless geek or an arrogant prodigy; and sometimes a gentle idealist. When he tells you the story of his life, it’s hard to figure out whether he has been very lucky or very unlucky up to now.

His hair is dark, his skin almost too pale. When we first met him, he was rather pudgy. He claims to have gained weight between 2011 and 2014 because of the constant stress he was undergoing. Almost from the get-go, his virtual castle, Mt. Gox, had been the object of numerous hacking attacks.

Karpelès was born in Dijon, the capital of the department of Côte-d’Or, in the region of Burgundy. Dijon, incidentally, has been the mustard capital of the world since the thirteenth century. In 1866, Maurice Grey, a mustard maker with a special recipe involving white wine, went into partnership with Auguste Poupon, his financial patron, to establish Grey Poupon—for Americans, probably more famous than Dijon. It is certainly more famous than Mark Karpelès.

Karpelès’s mother and grandmother, his only family, called him Robert —Mark Marie Robert, officially—because his grandmother was also named Mark. Yes, that’s confusing, and it was confusing to Karpelès. “Robert” in Latin means “red bear,” and “Karpelès” in French means “castle in the sun.” Later, when he tried to think of a Japanese name for himself, advice from

Japanese friends made him decide on Youjou Kumakichi. *Youjou* refers to castles or fortresses, and *Kumakichi* to an anthropomorphic red bear character in a manga series who is sexually perverse and always getting arrested. Perhaps his friends were pulling his leg when they suggested the name. Or maybe Karpelès himself has an ironic, even prophetic, sense of humor.

About his father, almost nothing is known. Karpelès's birth certificate says as much. "Father: unknown."

"I have trouble remembering many of the things that happened to me in my life," he told us in French and accented English when we were interviewing him. "But at the start, even my birth was a huge pain, apparently. There were problems—I spent the first month of my life in the pediatric ward of the Clinique de Chenôve in Dijon. I was there because of a left kidney deficiency and a gastric complication."

Karpelès was a sickly child. It was no easy task for his mother, Anne Robert Karpelès, to raise him. She was a geologist who later became a real estate agent, working full-time to support the two of them, and sometimes having to leave him in his grandmother's care. Home for mother and son was in Dijon and Paris most of the time.

He was also a temperamental child. For instance, he convinced himself that he had tapeworms, and he checked the Internet for how to tackle them. One website declared that eating raw onions was the cure. These he ate for three solid days without telling his mother about it—until she couldn't help noticing.

Once, when he ran away from home—something he did often from an early age—the local gendarmerie found him at the Quai d'Orsay in Paris, where he explained that he'd run off because his mom had tried to force him to vacuum the living room.

Grandma Mark, on her side, would punish and torment him in a variety of strange ways. She once forced him to comb his hair ninety times on one side of the head, and she repeatedly said she wished he had been born a girl.

When Karpelès was three, he was given his first computer. It was a Sinclair ZX Spectrum, a home computer that was popular in France at the time. If you are one of those people who can remember when RadioShack sold the TRS-80, the Spectrum might be familiar: 128 kilobytes of RAM with a tape recorder. Karpelès started to create basic programs on it immediately. Between the ages of three and seven, he was supplied by his mother with computer games, which he would try to modify for fun. He would also dismantle household goods such as calculators to find out how they worked.

Karpelès was fascinated by his computer and felt very much at home coding. But being an exceptionally bright child had its downside. Public school didn't go well because he was so far ahead of the others in his class. His teacher scolded him for counting to a hundred when she was only teaching the class to count to thirty. He was already able to read and write at a young age.

Anne Karpelès says, "I wasn't happy with his teachers, who seemed more concerned about keeping everyone at the same level rather than helping Mark fulfill his potential." She put him in a school for gifted children, where classes were based on ability rather than age. Some classes would have six-year-olds studying alongside twelve-year-olds.

Private school, however, was expensive, and when she lost her job, with her son then ten, Anne had to put him back in public school. He didn't fit in and was bullied and considered strange. He retreated from the world and became depressed.

Sometimes he got into trouble. When he was arrested as a teenager for some minor offense, he asked the police as a joke to put him in handcuffs because he wanted to feel as if he was in a movie. Life to him was becoming unreal.

His worried mother decided to send him to the Prieuré de Binson, a Catholic boarding school in Marne, for the next five years of his secondary education, from about age eleven to sixteen. At seventeen, after failing his first year at the Lycée Claude Bernard, he announced that he'd had enough. Anne convinced him that if he spent two years at a vocational school in Paris he could get a degree in electrical engineering and earn a decent living. Thus he didn't go to university and study computer engineering as most French software developers usually do.

Karpelès didn't fit in at the vocational school either. He had issues with the tougher kids, and he was let down by people he thought were friends. Life in a tiny apartment shared with his mother gave him little privacy. He left home and decided to live on the streets. For weeks he wandered Paris, distributing flyers for a cybercafe in the neighborhood of Châtelet.

In France, where most apartment buildings have electronic doors that require codes to get in, Karpelès would simply watch people punching in their numbers and then let himself in at night. He found that he could usually get a decent night's sleep on the third floor of buildings with elevators, since almost no one would climb the stairs that far. "Wooden stairs were better than stone," he explains. The money he'd made during the day allowed him to buy food at night. He spent some of his meager funds in cybercafes.

It wasn’t very long before the police noticed him, and he was taken to a halfway house for homeless youths.

Karpelès stayed there for a few weeks and then found a roommate. This was L, a guy he met over the Internet who was a member of Manga4All, a Japanese comic book fan club.

During his forays on the Internet, Karpelès had become fascinated with Japan—at least the futuristic anime version of it he took to be reality. Anime is a form of animation originating in Japan that depicts fantastic characters in vibrant graphical backgrounds, often with fantasy, horror, or science fiction themes.

To him, Japan seemed like a place where he could finally fit in, where a cyber geek would be appreciated—if he ever got there. Japan and coding: these were the two things he knew he loved.

One of his other interests was playing Counter-Strike, an online multiplayer game in which the player acts as either a terrorist or a counterterrorist. Karpelès decided to e-mail the French host of the website where he played, Stéphane Portha, and ask him if he needed any development programmers.

The response was immediate and enthusiastic.

In October 2003, Linux Cyberjoueurs—the company owned by Stéphane Portha—formally hired Karpelès. According to Karpelès and other sources, Portha was a pioneer of cybersquatting on domain names ending with “.fr”—for France. Cybersquatting was once a popular moneymaking scheme in which people registered attractive domain names before reselling them at a higher price to interested parties. Karpelès worked as a technician responsible for creating programs and dealing with administrative tasks for the Linux-based system the company had in place.

For a short time, he was happy. He had a job and he was being paid to code.

“I stayed one year and nine months at that company,” Karpelès said. “But I had a few little issues with the owner. I left the job because I was tired of being exploited. I also wanted to work in another environment. I wanted to be freer.”

Portha was reluctant to let Karpelès leave. He had become an essential part of the operation. According to Karpelès, they had a huge argument when he announced he was quitting. Portha allegedly warned him, “If you leave this company, I will make your life a living hell.”

Karpelès, unwittingly or dim-wittingly, helped make that true. In the process of leaving the firm, he transferred some client data on his employer's servers to other servers in France and the United States. He also took over a domain name and redirected it to his own online address. Domain names in those days could be worth a lot. Karpelès offered Portha $1,667 for the domain name and the data he had diverted. Portha didn't accept the offer and instead filed a criminal complaint.

Karpelès was still only twenty years old when this occurred. He didn't seem to understand that there can be lasting problems after a commercial divorce. But, then again, he had always been a little out of touch with reality.

After Linux Cyberjoueurs, he worked at Fotovista, the operator of Pixmania, before moving to Israel, where he stayed nine months.

Selling computer services there was a good idea, since Israel is a center of technology. But terrorist attacks were not something he had planned for. When the nearby electric power plant was blown up, his web-hosting business was blocked for twelve hours. Twelve hours of downtime in cyberspace is a very long time. In almost a single day, his business went down in flames, metaphorically and almost literally.

Still, Karpelès was undeterred. "Failure can help you improve. You can't be successful if you don't keep trying."

In 2006, he returned to Paris with only one bag on his shoulder. All his other possessions were lost by the airline. For a few months, he lived in an apartment with an older friend and benefactor.

Then, thanks to someone he met online, Karpelès got a job at Nexway, a telecommunications firm known at the time as Téléchargement.fr. He started working there as a developer and rapidly moved up in the company. He left his friend's place, relocating to 78 Avenue Félix Faure, in Paris's 15th arrondissement.

"The work at Nexway was great, and most of my colleagues were nice. I had a good relationship with the CEO, even after I left the company. Of all the professional experiences I've had, those years with Nexway were the most enriching—on all levels."

Christophe Eblé, the chief architect there, formed a very positive opinion of him. In an interview he told us, "Mark is an extremely talented guy who has an unbelievable knowledge of almost every part of computer programming, analysis, and architecture. Mark is not only a talented programmer but a generous, thoughtful person as well!"

In the summer of 2007, Karpelès decided to make a trip to Japan with a

dozen friends. “Just before the journey, I invited everyone to my place and stuffed them with quiche and apple pie, on the pretext that I needed to empty my fridge before going away for a month.” It was one of the happiest times in his life.

The trip was organized by people who mostly went by their online names: “Hakkai,” “BombStrike,” “Ookami,” among others. Karpelès isn’t sure he can even remember their real names.

“We were divided into two groups. Mine stayed at a guesthouse in Saitama Prefecture, near Omiya. During this month in Japan, I traveled a good deal. I went to Nara, Kyoto, and Osaka. I visited almost all the temples in the region, including several historical spots. I walked a lot and took a hell of a lot of photos.”

When he returned to France, he continued his work at Nexway. But when his grandmother died in 2009, leaving him only her cat, he flew back to Tokyo, this time to work there. He’d arranged to have himself transferred to Nexway’s newly acquired Japanese company, known at the time as CoGen Media, now called Degica.

Karpelès arrived in Japan on June 18, 2009, with his cat, Tibane, and moved directly into a new apartment at Fleur Tsuzuki 102—his first address in Tokyo.

Karpelès loved Japan.

In particular, he liked the quality of life. He appreciated the fact that he could buy things in convenience stores twenty-four hours a day (rare in Paris, where on Sunday even the supermarkets often close early) and that all sectors of society, but notably the delivery and transportation systems, were extremely efficient.

“In Japan, you get your health insurance card in twenty minutes at City Hall. In France, it’s more complicated. I had to wait two hours to get one there.”

Both France and Japan have public health services that are relatively inexpensive and make basic health care affordable and easily accessible. This was another plus.

He also liked that if he left his laptop on a park bench, it was likely to be returned to him. In Paris, he said, people would simply have stolen it. (Of course, someone who leaves a laptop on a park bench expecting to retrieve it later may not be the best person to handle cybersecurity for a firm dealing in hundreds of millions of dollars, but that’s another story.)

Karpelès admired the Japanese for their politeness. It was this, he figured, that allowed him to integrate into their society, and he learned to respond to it in a similar way. Even on a crowded subway, people behaved considerately. In France, it was the exact opposite. “Every morning when I took the metro in central Paris, I took precautions, always keeping a tight hold of my bag.”

But the thing he liked most about the country was its anime culture. Akihabara, a mecca for electronic equipment, was also home to various spin-offs of that artificial world. One establishment he frequented was a café where the waitresses dressed as old-fashioned maids and addressed their customers as “Master.” Later on, he’d spend a lot of time at a less savory place near JR Akihabara Station, a step away from his favorite electronics store and a kebab shop he enjoyed.

A couple of months after his arrival, Karpelès quit his job at Nexway’s Japanese branch to create his own company, Tibanne Ltd., which was registered on October 29, 2009. He named the company after the cat (with only one *n*) he had inherited from his grandmother and taken with him to Japan.

The company provided web hosting and software development, which he took care of himself.[1] At the same time, he was hired on by other firms, including Degica, as a sort of cyber *ronin* (a lordless samurai).

In January 2011, Jed McCaleb, a programmer, entrepreneur, and the owner of mtgox.com —an online exchange for the card game Magic: The Gathering—approached Karpelès to see if he would like to buy his business. Founded seven months before, Mt. Gox was already a bitcoin enterprise.

It was a prospect that interested Karpelès because he had been fascinated by bitcoin since 2010 when William Waisse—a French friend and client of his based in Peru—asked if he could pay him for his services in bitcoin. It had occurred to Karpelès that, with the right platform, he could create a bitcoin exchange that would make buying, selling, and using the virtual currency much easier.

Karpelès was in. He acquired Mt. Gox while he was still working in Degica’s office. The purchase seemed to be a good deal to him at the time. “What I’d bought, though,” he told us later, “was a lot like a treasure chest with a time bomb inside.”

To him, bitcoin was an amazing innovation, which he felt could become a worldwide currency if there were the right trading center for it. In the rudimentary system that had been set up to buy, sell, and exchange Magic: The Gathering trading cards, he saw great potential.

Of course, others would later joke, “Take a website made for exchanging playing cards and turn it into a platform trading millions’ worth of virtual currency—gee, what could possibly go wrong?”

However, 2011 would turn out to be a tough year for the exchange, for Karpelès, and everybody in Japan. On March 11, the country was rocked by a colossal earthquake, a tsunami, and the triple meltdown of the nuclear power plants in Fukushima Prefecture. It seemed like the end of the world for a few days. The government kept silent, but Naoto Kan, the prime minister at the time, feared that the badly damaged Fukushima reactors would go off one after another, making it necessary to evacuate Tokyo.[2] The Japanese economy itself seemed to be on the brink of disaster.

Meanwhile, in response, the French embassy began organizing special flights back home for French citizens. Many took them during the panic that ensued due to the Japanese government’s apparent inability to contain the nuclear crisis. Karpelès was not invited to fly, since he was not registered at the embassy. Repeated advice from various people made him decide to go to Osaka for several days, but it wasn’t practical for work.

In due course, the work with Mt. Gox started to expand. Soon Karpelès had no time for anything else. Around June 2011, he hired his first full-time employee, Adam Turner, a.k.a. Ashley Barr. It was necessary to have a support team in place, and Karpelès decided to rent an office on the fifteenth floor of the Cerulean Tower in Shibuya, a fashionable district in central Tokyo. “The advantage of being there was that we would have the use of meeting rooms, somewhere to chill out, and an office with everything we needed—tables, chairs, Internet. Rapidly, the support team led by Adam grew to include three new employees: C, N, and J.”

In June that year, the site was hacked. It had been hacked before, but this time the intrusion was much more serious. Someone had gotten into the system and manipulated the account balances. The hacker—or hackers—was able to sell off all the bitcoins from people’s accounts, drive the market price down, and withdraw a few thousand bitcoins themselves.

But Karpelès was able to keep it all together.

“At the time, Mt. Gox’s withdrawal limit of $1,000 a day applied to bitcoins at the real time rate, so hackers wanted the rate to drop to be able to withdraw more,” he explained. “To do so, however, they crashed the Mt. Gox system and were unable to withdraw all our funds as part of their plan, because Neo futur [a friend in Peru] woke me up in the middle of the night and I was able to take the site off-line to stop the attack.”

The damage could have been bigger if the hacker had access to anti–

money laundering certified accounts. These were accounts in which the user had provided verifiable ID and thus was allowed to withdraw up to $10,000 a day.

The setback wasn't permanent. Mt. Gox was highly profitable, and Karpelès had more money than he ever had before.

He was also creating a family.

He had married a Japanese woman, Kyoko, who worked at Degica, and together they had a child, Pierre. She left him in 2013 because, people close to her said, Karpelès was too busy with his work and never spent enough time with her. When she left Japan, she took Pierre with her. Karpelès became depressed. "It's only after you've lost someone that you realize how important that person was." It's a commonplace sentiment but no less true for its unoriginality.

It was not long after she left that he started to see a wide variety of women, some willing companions, some he paid to be with him. In Japan, where all sexual services besides vaginal penetration can be legally bought and sold, his behavior wasn't so far outside the social norm. According to reports in the mainstream Japanese press, he spent over $170,000 of company money on credit card payments and women. On October 29, 2015, the *Yomiuri Shimbun*, the *Sankei Shimbun*, Asahi Broadcasting Corporation, and other major media all ran stories quoting the police saying that Karpelès had paid money to prostitutes.

Karpelès often told his friends that he was sad that his wife had abandoned him and he was lonely—and horny.

He was, however, well known for his fondness for pussy—both the feline kind and the female one. Sometimes those fondnesses overlapped. He made at least one girlfriend dress in a fluffy cat costume before sex.

According to former lovers and friends, his sexual appetite is immense and his preferences unusual, leaning toward light S and M play, sometimes B and D. They say he has the ability to ejaculate large amounts of sperm at some velocity. While engaged in intercourse with the "cat woman"—a lady dressed in a whole-body kitten costume—apparently he had an orgasm that shot his sperm across the room and left a white stain on the side of his luxurious ¥6 million yen bed.

But Karpelès's stained furniture would become the least of his worries as Mt. Gox grew. The story of its development into the epicenter of the bitcoin universe, and then a black hole, is worthy of a chapter on its own. There were forces at play that helped turn the company from a mecca into a

disaster area—forces that included people we usually think of as the “good guys.” But before discussing this in more detail, we need to take a slight detour—a detour down the Silk Road.

CHAPTER THREE

THE SILK ROAD

The first bitcoin purchase may have been for pizza, but it didn't take long for people to find out that bitcoin could also be used for guns, drugs, child porn, and other things you might not want people to know you are buying or selling. When you combine bitcoin with the "invisible Internet," a.k.a. the Darknet, or deep web, and a privacy-protection Tor Browser, as well as an online catalog of illegal goods, you have all the ingredients for an underground marketplace.[1]

That marketplace was, and still is, called Silk Road.

In the vocabulary of modern-age Internet users, Silk Road is not the ancient trade and intercultural route, six thousand kilometers long, between East and West. No, it was the first online black market and platform for buying and selling illegal goods. It started functioning in February 2011 and was once the largest dark-web marketplace for illegal drugs and other services.[2] The only thing you had to do to access it was download Tor—an acronym for The Onion Router—and search places like the Hidden Wiki, a secretive deep-web page that references anonymous ".onion" sites, including Silk Road. These .onion sites are only visible when you use Tor.

The hard-to-find .onion sites are part of the deep web—a corner of the web that isn't open to search engines like Google and normal users. It is full of sites for people intent on preserving their privacy online.

Naturally, the seedy side of the Internet can be found here, including some elaborate scams and hoax websites masquerading as underground businesses. But there are also beneficial uses for .onion sites. For example, a whistle-blower or political activist in Iran or Egypt, or even China, can use them to upload and share secret data or communicate to the outside world without fear of reprisal.

Oddly enough, Tor was originally developed by the United States Naval Research Laboratory and mostly financed by the US government. It is a superb tool for protecting user privacy. Tor is available free online and acts as a "proxy." According to the Tor Project website, its software protects users by bouncing their communications around a distributed network of relays run by volunteers all around the world. It can prevent third parties watching the user's Internet connection from learning what sites they visit; it prevents the sites they visit from learning the user's physical location; and it lets the user access sites that are blocked in their own country.

The Tor Browser, which is based on Firefox, lets anyone use Tor on Windows, Mac, or Linux operating systems without actually needing to install any software. You can run it off a USB flash drive that comes with a preconfigured web browser to protect your anonymity, and it is self-contained and portable. In short, you can take Tor with you into any net café in the world and surf with impunity. Or you can do it from home. And you might find yourself browsing topics and sites like "Rent-a-Hacker," "Create encrypted self-destructive notes," "Drug Market," "US Fake ID Store," "Hitman Network," "Counterfeit USD," "OnionWallet—Anonymous Bitcoin Wallet and Bitcoin Laundry," "PedoEmpire," or "Anonymous, safe, secure, crowd-funded assassinations."

Incidentally, Mark Karpelès's company hosted a website called silkroadmarket.org, which explained how to get started with Tor and reach Silk Road. The connection was made under a false name, and the fees paid in bitcoins to Karpelès. He had become a part of Silk Road without really knowing it.

While Silk Road was—and is, in the form of Silk Road 3.0—a marketplace for illicit items, there is a strange honesty to the whole enterprise. Vendors are unlikely to cheat because the site runs on a reputation-based trading system. If you sell a product and don't deliver, your rating will plummet. Acquire a bad reputation, and your business is not sustainable.

The website Gawker first reported on the existence of Silk Road in a June 2011 article. "'Our community is amazing,' Silk Road's anonymous administrator, known on forums as 'Silk Road,' told us in an email. 'They are generally bright, honest and fair people, very understanding, and willing to cooperate with each other.'"[3]

They say there's no honor among thieves, but with Silk Road vendors there is a healthy respect for reputation that borders on it.

Of course, some things, like fake passports, are probably hard to rate, since if you fail when using one, you may not have access to a computer to write a review for a few years.

Vendors feel comfortable selling illegal goods on Silk Road because the police can't easily track them. Silk Road does not accept credit cards, PayPal, or any other form of payment that can be traced or blocked. The only currency good there is bitcoin. The trouble is, bitcoin is not as anonymous as Silk Road users would like to believe, since transactions are recorded in a public log—the blockchain. The identities of all users on the blockchain remain anonymous, but the flow of bitcoins can be tracked, giving investigators a way in.

Silk Road really arrived on the scene around New Year's Day 2011. An unknown individual who used the pseudonym "altoid" started to advertise the hidden Tor service on Internet forums such as bitcointalk.org. He painted a portrait of a black-market eBay where you could find almost anything you wanted.

About two months later, a message on BitcoinTalk announced that Silk Road had been operational for a few weeks with several buyers and sellers. By April, Silk Road hit a thousand users, and by summer the site went mainstream after *Gawker* published its article on June 1. In that piece, investigative reporter Adrian Chen explained how Silk Road allowed people to buy and sell illegal goods safely, using bitcoins for transactions.

A few days after the article came out, Democratic Senators Charles E. Schumer and Joe Manchin sent a letter to the Drug Enforcement Agency in which they expressed serious concern and asked the authorities to take immediate action to shut down the site. According to his partisans and family, that's around the time that Ross Ulbricht, a physics graduate from the University of Texas at Dallas, relinquished control of the site to his successors. Ulbricht was twenty-seven at the time.

On June 18, the official Silk Road forum appeared on Tor, and Ulbricht allegedly posted his first comment there under the username "Silk Road." Then in October the BitcoinTalk user altoid posted a job ad for an IT professional in the bitcoin community. It was a job for Silk Road, and people interested were asked to write to rossulbricht@gmail.com.

Ulbricht was far from being the Professor Moriarty of the dark web. On February 5, 2012, the user Silk Road announced that his new name would be "Dread Pirate Roberts." This alias was taken from the popular book and film *The Princess Bride*, written by William Goldman.

According to the criminal indictment filed against Ulbricht, Silk Road wrote:

> I need an identity separate from the site and the enterprise of which I am now only a part. I need a name.
>
> Drum roll please . . . My new name is: Dread Pirate Roberts."[4]

Thus Dread Pirate Roberts was born—and the police began looking for him immediately. They didn't, however, shut down the site just then. Maybe they couldn't.

Silk Road at the time was a modestly successful venture. By March 2013, the site listed ten thousand items for sale, seven thousand of which were

drugs, including cannabis, MDMA, and heroin. The FBI closed Silk Road down in October that year. The agency asserts that by the time it was closed down, Silk Road boasted an estimated nine hundred thousand users and had generated more than $1 billion in sales.

The FBI may have shut it down, but they couldn't destroy the marketplace. Today, it still operates in a revised version called Silk Road 3.0, with complementary websites telling you how to access the new and allegedly even more secure outfit. The FBI doesn't like to talk about it. It makes closing down Silk Road look like fighting the Hydra and losing.

And how was Ross Ulbricht arrested?

It took years of work. Homeland Security Special Agent Jared Der-Yeghiayan first began investigating Silk Road while he was working at Chicago's O'Hare International Airport in 2012. He noticed that several seized drug shipments could be matched up with photos, descriptions, and "shipped from" locations as advertised on Silk Road. This is how the investigation began.

Der-Yeghiayan put together a team at the Department of Homeland Security (DHS) and set up an undercover Silk Road account under the username "dripsofacid." His team then transferred roughly $7,000 to Mt. Gox in exchange for 27.27 BTC on April 5, 2013, and began to buy things.

Over the course of the investigation, the special agent logged thousands of hours on the Internet, and his team made nearly fifty undercover purchases. Surprisingly, most of the dealers were "honest"—the drugs shipped were tested and found to be pretty much as advertised in roughly nine out of ten cases.

Der-Yeghiayan worked himself further into the network under the name "Cirrus," becoming responsible for assisting clients and moderating inappropriate comments on the site's forums. He was paid the equivalent of a $1,000 a week in bitcoin, proving that crime may not pay but customer support jobs almost always do.

Homeland Security Investigations, the IRS, the DEA, and the FBI joined forces to home in on whoever was masterminding this covert marketplace. Their mission was named "Operation: Marco Polo" after the famous explorer of the ancient Silk Road.

To arrest their prime suspect, Ross Ulbricht, the idea was to get him somewhere out in the open and initiate an online chat with Dread Pirate Roberts, then grab his computer in a nonencrypted state to verify that it contained the same exchange. If Ulbricht managed to close down his laptop

before it was taken from him, it might be impossible to prove that he and Dread Pirate Roberts were one and the same person.

In the afternoon of October 1, 2013, Ross Ulbricht, wearing a T-shirt and a jacket, was sitting in the science fiction section of the Glen Park Library in San Francisco, taking advantage of the public Internet. Der-Yeghiayan was sitting opposite the library, in Café Bello. At 3:08 p.m., when Dread Pirate Roberts connected on the Silk Road chat site, Der-Yeghiayan asked him to verify a message on another forum. “OK, which post?” was the reply. At that moment, the special agent gave the green light for an arrest. In the library, a woman wearing a yellow raincoat suddenly yelled, “I’m tired of you!” and while Ulbricht was looking at her, his laptop was snatched out of his hands. The woman in the raincoat was an undercover FBI agent who had told the librarians about the planned operation. There were half a dozen agents already in the library. They forced the man onto the floor and put him in handcuffs. Ulbricht submitted without resistance and without emotion. He was arrested on a number of charges, including narcotics trafficking and computer hacking.

Predictably, the FBI took most of the public credit. One of the reasons why many special agents in other federal law enforcement agencies are not fond of “the bureau” is that many investigations play out this way—other agencies do the bulk of the work, and the FBI gets the kudos. The laziness of the press also aggravates this problem, as they automatically tend to credit the FBI rather than lesser-known agencies.

The trial was held more than a year later but lasted only three short weeks. Ulbricht’s defense attorneys tried to argue that he was the fall guy—not the real Dread Pirate Roberts—based on the fact that US law enforcement had focused on other suspects for two years. His attorneys also argued that while Ulbricht had created Silk Road, he had long since handed it over to another party who was actually running it.

Among the other suspects, apparently, was Mark Karpelès. Yes, the hapless MagicalTux.

Ulbricht initially pleaded not guilty to seven charges, including narcotics trafficking, criminal enterprise, computer hacking, and money laundering, according to media reports.

Before he was sentenced, the parents of a number of Silk Road users who had died from overdoses of substances allegedly purchased on Silk Road told the court that their children would never have died if the site hadn’t existed.

Summing up, the judge told him, “The stated purpose [of Silk Road] was to be beyond the law. In the world you created over time, democracy

didn’t exist. You were captain of the ship, the Dread Pirate Roberts. You made your own laws.”

In a New York courtroom on May 29, 2015, Ulbricht was sentenced to life in prison without parole.

The accused was polite and well educated, and even his closest friends were astonished to learn of his activities in what seemed like a parallel world. How could this guy who got along with everyone (albeit a little clumsily with women) end up becoming his country’s public enemy number one in such a short period of time?

Although the bulk of the evidence suggests that Ulbricht was indeed the creator of Silk Road and Dread Pirate Roberts, his supporters contend he is innocent. Some in the bitcoin community believe that even if he were Dread Pirate Roberts, he had committed no real crime. One such defender is Roger Ver, pioneering investor in bitcoin, who says he donated around $165,000 to Ulbricht’s legal defense. Ver is an ex-convict himself, as well as a former US citizen. He is also known as the Bitcoin Jesus because of his staunch advocacy for the virtual currency and his zeal for spreading the “good news” about it.

“My point of view is that none of this would ever have happened if it wasn’t for the government’s insane war on drugs. If governments respected each individual’s self-identity, and allowed them to actually control their own bodies, drugs like these would be sold by CVS, Walgreens, or Amazon.com,” said Ver during interviews we conducted with him. “There would never have been a need for Silk Road, or any of the problems related to it.”

Ver believes that Ulbricht was simply helping people get what they want out of life. In our interviews with him, he emphasized that Ulbricht wasn’t an active participant in criminal activity. “He wasn’t using any violence on anybody. The indictment doesn’t say Ross smuggled drugs; what he did [was provide] a platform that allowed other people to buy and sell them. By the same logic, should we charge cell-phone companies because a lot of drug deals have been done by people talking on cell phones? What about the Internet as a whole? Or e-mail providers? All these services enable people to buy and sell and smuggle narcotics.”

While the Silk Road investigation was going on, another kind of criminal activity using bitcoins began to proliferate in cyberspace. The popularity of bitcoin with criminals has an obvious cause: it’s held in a digital wallet that isn’t registered with any government or financial authority, and it can easily be swapped for other currencies through exchanges such as bitFlyer, OKCoin, Bitfinex, BTCChina, Bitstamp, or any of a dozen more now operating in

many countries. Ransoms thus become virtually untraceable, and because bitcoin transactions are made to be irreversible, the money can't ever be retrieved.

About the time that Silk Road was being closed down, hackers around the world managed to enter and take over a vast number of computers—sometimes taking down public websites—and hold them for virtual ransom. Their victims were random computer users, financial firms, and even police departments. Usually the only way to regain control of the computer was to pay the criminals a ransom in bitcoins, sometimes as much as $20,000. For firms or individuals with valuable files saved on their computers, sometimes with no backups, the threat of losing access to data was obviously very serious.

One recent case involved a criminal or group of criminals who went by the name DD4BC. The raider threatened to shake down the websites of several financial companies with overwhelming message traffic unless a bitcoin payment was made.

The most successful of the "ransomware" operations was CryptoLocker, which flourished from late summer of 2013 until early summer 2014. International law enforcement identified the mastermind behind it as a thirty-year-old Russian named Evgeniy Bogachev. In a June 2014 report, the Department of Justice cited security research suggesting that as many as 234,000 computers had been compromised by the malware. Bogachev and a group of cyberpirates believed to be based in Russia and Ukraine were supposedly running the CryptoLocker scam and reportedly made about $16.5 million in bitcoins in a little more than a month in 2015.

Roger Ver was the victim of one of these new cyberpirates while he was attending the Singapore Coin Congress in May 2014. Somebody hacked into all his old e-mail addresses, a forgotten Facebook account, and Skype, and they asked him to pay a ransom of $20,000 in bitcoins, threatening to "ruin him" and harass his family in the United States by sending a SWAT team to his mother's house—possible because the hacker had Ver's family's address.

This malevolent practice is known as "swatting" in the cybersphere. Hackers and gamers anonymously report false shootings, hostage situations, and other crimes so that SWAT teams and/or other paramilitary police units will raid the homes of the victims—often with no warning and breaking down their door in the process.

"All I want is some bitcoin and I will leave you alone," said Ver's hacker, who was using the Skype account "nitrous" and the screen name "Savaged" when he wrote to Ver. The man bragged that his take from Ver

included his social security number and other private data. He promised to erase the stolen information and stop infiltrating Ver's cyberlife only if Ver paid up.

Just thirty minutes after the hacker contacted him, Ver decided to employ the tactic used by Mel Gibson's character in the movie *Ransom*. He would use the Internet community to flush him out.

He had an unspoken message for the pirate: *Do not fuck with the Bitcoin Jesus. He is a vengeful God.*

Rather than hand over any bitcoins, he posted ads on bitcoinbountyhunter.com, Facebook, and Twitter. He offered the same amount the hacker had asked of him ($20,000 worth of bitcoins) as bounty on the hacker instead of paying the ransom, and he mentioned the Skype username nitrous.

Ver's post was retweeted by notable Internet figures, and thousands around the world were inspired to help find the culprit by pooling information. Meanwhile, Ver rallied his bitcoin friends from Tokyo, including Jason Maurice, who at the time was the chief white hat hacker at security consultancy Wiz Technologies.[5] Both Ver and Maurice happened to be staying at the same hotel in Singapore where the conference was being held. Maurice immediately dropped whatever he was doing to start working with Ver. Together they began limiting the damage by identifying accounts that hadn't been compromised and locking them down more securely.

The hacker was trying to hijack Ver's domain names and primary e-mail account by using information obtained from his old Hotmail account. Maurice fought hard to outmaneuver him.

Someone on Skype known as "TGOD" even claimed he knew the hacker personally and had a grudge against him, but he wouldn't give him away until he got paid all or part of the 37.6 BTC reward in advance.

About an hour after the incident started, the hacker, irritated by the lack of a response, threatened Ver by writing that he would hit him "10,000 times harder" than he had originally planned.

This didn't work.

Then the pirate switched tack. He said he was demanding the money to help pay for his mother's $15,000 "transplant procedure," but he gave no concrete details. Was it a liver transplant? A heart transplant? It didn't matter. Ver was unsympathetic. He sent the hacker a link to his Facebook post offering the bitcoin bounty.

At this point, the hacker seemed to panic, sending frantic apologies and claiming he was representing another person.

> Sir, I am sincerely sorry I am just a middleman I was being told what to tell you.
>
> I was seriously being told what to tell you by someone else I don't even know what's going on
>
> Please stop I am so sorry I told him that you are now going to have me killed over something he made me do I didn't even do this it was someone else

But Ver didn't let it go. He replied by reminding the hacker he could earn 37.6 BTC by turning in the real offender.

The response was:

> man that isnt even me this is so fucked up i got myself in this situation
>
> You dont know the stuff he makes me do he did this to me before . . .

The hacker vanished after saying he would probably turn himself in to the authorities and notifying Ver that he had deleted the Hotmail account and reset all passwords to his other accounts. He also tried to delete the threats he had made on Skype, but Maurice got the chat logs just in time and posted the contents online.

The actual hacker's identity hasn't yet been ascertained, but a couple of facts are certain: it wasn't Ross Ulbricht, and Ver will still pay 37.6 BTC to whoever helps get the actual culprit arrested.

Ver still doesn't see bitcoin as the currency of criminals, even after this episode. His support for Ulbricht also remains unshaken.

Is there any merit to Ross Ulbricht's defense? There may be. There is certainly some doubt that he received a fair trial.

A curious interview exists with another man claiming to be the real Dread Pirate Roberts. It was published in *Forbes* in August 2013 as "An Interview with a Digital Drug Lord: The Silk Road's Dread Pirate Roberts (Q&A)." A month earlier, the journalist had put a series of questions to this man through Tor and the Silk Road messaging system. When asked why he had started Silk Road, his answer was: "I didn't . . . my predecessor did. From what I understand, it was an original idea to combine Bitcoin and Tor to create an anonymous market. Everything was in place, he just put the pieces

together.”[6]

Was this Ross Ulbricht already trying to distance himself from his own work? Had Silk Road taken on a life of its own after Ulbricht had created it, much as Satoshi Nakamoto had created bitcoin and then seemingly vanished into cyberspace?

Who is the real Dread Pirate Roberts?

Some investigators believe he was still operating even after Ulbricht’s arrest. Some seem to share doubts as to whether they really had the right man. Some believe that the real Dread Pirate Roberts was an insider at Mt. Gox.

CHAPTER FOUR

THE MAGICAL MT. GOX

Let's talk about Mt. Gox again.

From the outset, Karpelès was a computer geek and math wiz, not a libertarian, politician, or financier, so he was never well informed about the political and financial aspects of bitcoin. It was the technology behind it that motivated and intrigued him.

As you may remember, Karpelès learned about bitcoin in 2010, when he was providing web-hosting services in Tokyo and William Waisse, a French client of his in Peru, asked if he could pay in bitcoin. Karpelès agreed, and he started to do some research on the currency. He thought the whole concept behind bitcoin was well thought out. There were a lot of technical problems that made for interesting challenges as a programmer. The beauty of the challenges themselves was sufficiently motivating for him (and others) to become involved.

Mt. Gox was the first successful bitcoin exchange that could cope with a massive number of users anywhere in the world. At its peak, it handled 80 percent of all bitcoin transactions.

The company became famous under Karpelès's ownership, but it was an American named Jed McCaleb who started the site.[1] In 2007 he bought the domain mtgox.com, which stood for "Magic: The Gathering Online eXchange." He tried to turn it into an online card-trading site but shut it down after only a few months for reasons yet unclear.

McCaleb was intrigued by bitcoin as soon as he heard about it. According to court documents, e-mails, and associates, he was eager to buy the currency but found it unreasonably difficult. He turned his long-dormant mtgox.com into a bitcoin exchange in July 2010, implementing balances, deposits, and withdrawals on the basis of his order-matching system already in place from his card exchange. In other words, he added bitcoins as another commodity that could be exchanged online. The new site allowed trading between bitcoin and local currencies, and it was the first of its kind. (Incidentally, McCaleb is another individual thought to be the elusive Satoshi Nakamoto. It seems unlikely.)

Mt. Gox soon became more than McCaleb had bargained for. In fact, it became so popular that it took up all his time. He began looking for someone to help him manage the site and found MagicalTux—Mark Karpelès. McCaleb turned to him for advice and then sold the site to him for almost

nothing.

Karpelès thought the conditions were favorable. There was no need for an initial payment, just an agreement to share 50 percent of the profits for six months, with McCaleb retaining 12 percent of the company. Since Karpelès had already moved to Tokyo, the new Mt. Gox Co. Ltd. was based there as a subsidiary of Tibanne.

Karpelès should have paid closer attention to the fine print. The main flaw in the deal was that the site had already suffered frequent bitcoin and monetary theft. Perhaps he didn't understand how that would affect him. According to some sources, a bitcoin theft even occurred on the day Karpelès acquired the site.

McCaleb wanted the matter kept under wraps and convinced Karpelès to sign a nondisclosure agreement. It's possible more details may be released during Karpelès's trial, but sources close to the Japanese police investigation believe that more than 80,000 BTC may already have been stolen by the time McCaleb sold Mt. Gox. Internal Mt. Gox documents we obtained corroborate this.

Within a few months of its acquisition, the company went from having three thousand client accounts to sixty thousand.

On June 9, 2011, bitcoin skyrocketed, peaking at a new high of $31.91 on Mt. Gox. This drew unwanted attention.

On June 18, exactly two years after Karpelès settled down in Japan, and fourteen weeks after the earthquake-tsunami–nuclear meltdown, the first serious setbacks with Mt. Gox occurred. Karpelès was woken up at 3:00 a.m. that day. William Waisse, a.k.a. "Neo futur," called him on his cell phone to say that there were problems. Quickly, Karpelès confirmed what Neo futur was saying, and within seconds he had shut down the entire system.

Hackers had always wanted to invade the exchange. Some had tried (and sometimes managed) to create denial-of-service (DOS) situations. McCaleb had reportedly put a fail-safe system in place so that any missing coins would be replaced by missing fiat money, the value of which was less likely to change. But that initial cybertheft of 80,000 BTC began a spiral of trouble that may well have led to the firm's eventual financial collapse.

In May 2016, Jake Adelstein and Nathalie Stucky, the authors of this book, were sent internal e-mails, contracts, and other documents relating to the implosion of Mt. Gox. All the documents were printed copies, and the envelope had no return address but appeared to have been posted from the Kasumigaseki district in Tokyo. Along with information provided by a former

employee of bitcoin who had handled accounting for the firm, the documents reveal previously unreported details about how Mt. Gox failed and why. We set about verifying the e-mails with Karpelès's lawyer—Nobuyasu Ogata—and former employees, as well as sources in law enforcement.

This material, which included correspondence between Mark Karpelès and Jed McCaleb, suggests that Mt. Gox was plagued with problems from its earliest days, before Karpelès was involved. The documents, including the e-mails, were submitted to the Japanese courts as evidence for the Mt. Gox trial, which finally began in Japan in July of 2017 after nearly two years of delays.

In an e-mail dated January 18, 2011, the year Karpelès was first approached about buying the site, McCaleb wrote:

> Hi Mark,
> Please keep all this confidential I don't want to start a panic and I'm not sure I'll do it yet but I'm thinking I might try to sell mtgox. I just have these other projects I would like to devote more time to.
>
> Would you be interested? It could be very little up front and just a payout based on revenue or something.
>
> There is also an investment group that wants to fund mtgox. Probably around $150k. So you could most likely take it over with some cash.
>
> Let me know
>
> Thanks,
>
> Jed

When Karpelès agreed to purchase the company in February, he signed an agreement that included the unusual statement that "the Seller is uncertain if mtgox.com is compliant or not with any applicable US code or statute, or law of any country." It also included a clause saying, "Buyer agrees to indemnify Seller against any legal action that is taken against Buyer or Seller with regards to mtgox.com or anything acquired under this agreement."

An e-mail of April that year, relating to the missing bitcoins, was probably the beginning of Karpelès's nightmare:

> From: Jed McCaleb <jed@mtgox.com>
> Date: 2011/04/28 22:33
> To: Mark Karpeles <admin@mtgox.com>
>
> I can't tell how big an issue it will be to be short 80k BTC if the price goes to $100 or something. That is quite a bit to owe at that point but mtgox should have made a ton of BTC getting to there.

There is also still the fact that the BTC balance will probably never fall below 80k. So maybe you don’t really need to worry about it.

There are 3 solutions I have thought of:

- Slowly buy more BTC with the USD that Gox Bot has. Hopefully you would fill up the loss before the price got out of hand.

- Buy a big chunk of BTC (really just moving the BTC debt to the USD side)[.] If BTC goes up this is a huge win. Problem is there isn’t enough BTC for sale on mtgox. Maybe you could find someone on the forum to do it?

- Get those crystal island people to invest. They have 200+ BTC so they could fill in the gap.

Maybe you could just mine it ?

We have tried to reach Jed McCaleb for several months both through his e-mail and social media accounts, but he has not responded.

What is clear now is that the missing bitcoins might have been a small problem when Karpelès bought the company, but his own success would make the lost amount a huge burden.

In the beginning of April 2011, 80,000 BTC was worth approximately $62,400.

Maybe Karpelès went along with the suggested advice and figured he could make it back as he went along. But luck was not on his side. While he may have been trying to fill the hole, the price of bitcoins kept rising. By June 2, the value of the missing bitcoins had jumped to over $800,000.

Unfortunately for Karpelès, he had signed a punitive nondisclosure agreement that left him unable to discuss the loss, and he faced the Sisyphean task of recovering the missing bitcoins on his own—a problem that became greater by the day and sometimes by the hour as the value of bitcoins rose. Jed McCaleb must have been relieved to have the shortfall off his back.

It was not clear how the June 18, 2011, hack occurred, but investigators believe that hackers might have gained access to McCaleb’s administrator account, which was still active.

Karpelès’s reaction was to move the majority of the bitcoins off-line into what is called “cold storage,” placing them in safety-deposit boxes dispersed among various banks in Tokyo. He only left enough online to make sure transactions could be carried out.

The episode made him increasingly paranoid about hackers—almost obsessively so.

The man in charge of accounting at Mt. Gox says he urged Karpelès several times to reconcile the BTC balance, the online balance, and the cash balance, saying he needed to know where the money was. “But Karpelès said it was *mendokusai* [a pain in the ass]. He claimed it was difficult and risky, because you’d have to put the cold-storage bitcoins in a hot wallet [online], which made them more vulnerable to cyber predators.” Karpelès thought they were safer in cold wallets.

Virtual money was temporarily becoming paper money, and there were masses of it. The chief accountant understood Karpelès’s concern from a security perspective, yet he still felt that not reconciling the accounts was unwise. “But it’s his company,” he told us in an interview. “I thought, he’s the CEO, so I said okay.”

Some former employees say that Karpelès might have made it all work. Later, however, the freezing and seizure of $5 million of Mt. Gox funds in May 2013 was another huge blow, cutting into the firm’s operating reserves and probably signaling the beginning of the end. One accountant remembers: “The first time I realized we’d lost a pile was sometime in early February 2014, when Mark called me into his office and said, ‘There is a chance that Mt. Gox might have to file for bankruptcy.’ And he asked me to go to the law firm Baker & McKenzie the next day to discuss it with them.” Apparently, Karpelès was eerily calm at the time—but he was always that way. “He was always smiling. He could look you in the eye and probably tell you, ‘Your socks don’t match,’ or ‘Oh, the office is on fire and we’d better leave before we burn to death’ and he would have the same strange smile on his face.”

Was Karpelès a con man, a victim, a fall guy, or all of the above? Only the outcome of his trial will tell. But one thing is certain: he bought a company already missing tens of thousands of bitcoins, and within a few months of his taking over, due to the rapid rise in their value, Mt. Gox was in the hole for close to a million dollars.

Did the thief who took them take hundreds of thousands—worth hundreds of millions of dollars—more? Someone did, in what may well be the heist of the century.

The June 18 hack seems to have been the most significant theft at Mt. Gox that summer, but there were many minor security breaches in the same period. Around the time of the June 18, 2011, hack, it appeared that McCaleb’s administrative account had been compromised, and an unconfirmed amount of about 20,000 BTC was stolen. Trading was halted for

a week while the breach was resolved. That same month, the Mt. Gox user table was leaked. This contained thousands of usernames, e-mail addresses, and password hashes. Some clients in the leaked database had used the same usernames on MyBitcoin, a then-popular bitcoin wallet, and had their passwords cracked. Six hundred of them had their balances stolen. It was a fiscal disaster.

Mt. Gox was able to recover from this by offering a public apology and reimbursing the lost funds. But they wouldn't be able to make the same promise of reimbursement again.

The hacks in June were a wake-up call for Karpelès. The first thing he did was put Mt. Gox's bitcoins in what he believed was a safe place.

> I decided to use my portable PC running on Linux, due to the fact that it was shut down most of the time, which made it a more difficult target than a server. Then I encrypted the bitcoin wallet and put it on Dropbox to make sure that the bitcoins wouldn't be lost, even if the portable computer's hard disk died.
>
> Once I arrived at the office, and after posting an announcement on the site, I was finally able to understand what had happened. A hacker had created two accounts, credited a huge sum of bitcoins and US dollars, and artificially increased the sum in one of the accounts. He then tried to withdraw the bitcoins, but without success because I'd deactivated a routine that Jed had set up, and limited withdrawals to $1,000 per day at the current rate. The pirate then decided to force the rate down by selling a lot of bitcoins in order to be able to withdraw more. This caused an overload on the system, which made it impossible for him to withdraw the money before I woke up and shut down the entire system.
>
> Later on, I was able to discover that the hacker had used a flaw of the type called "SQL Injection" to get a list of the Mt. Gox users including Jed McCaleb's administrative account and use it to modify the balance in Jed's account. I didn't know if there were other flaws in it, so I decided to reopen the site with a new, clean system. This meant that I had to rewrite the code from scratch, and quickly. I also decided that everything that happened during the hack and after would be blotted out, treating it as if it had never occurred.

All this was known as the "Mt. Gox rollback" among Mt. Gox users at the time. Karpelès rewrote almost all the code used to run the website and

exchange. McCaleb's still-existent account had certain administrative privileges, and these were also changed.

Three individuals came to help Karpelès deal with the hack: the Bitcoin Jesus (Roger Ver), Jesse Powell, and one other cyberhero. Powell is the cofounder and CEO of Kraken, a leading San Francisco–based bitcoin exchange. These people provided a support team while Karpelès was trying to put a new Mt. Gox together that would go live as quickly as possible, even if it meant working night and day and during weekends.

"Conditions in the office with all these people around made it very hard to concentrate," Karpelès recalled. "So it took a while to rebuild Mt. Gox. I created a system to allow users to recover control over their accounts, which worked pretty well. In fact, to my knowledge, Mt. Gox is the only bitcoin exchange to have successfully recovered from such an attack. Later, in August 2011, a system of 'cold wallets' was put in place in order to increase security. Around that time, we also reformed the original bitcoin core software."

But Karpelès failed to solve the problems, according to experts at WizSec, a bitcoin security firm established in 2014 in Tokyo by former Mt. Gox creditors. They launched an independent investigation in early 2014, drawing on as many sources of information as possible, including transaction data leaked by hackers and interviews with ex-employees, in an attempt to reconstruct relevant parts of Mt. Gox's database. Bitcoin addresses, deposits, and withdrawals were matched against the blockchain to detect any irregularities.

Their conclusion was that Mt. Gox's system had been compromised by a hacker or hackers who slowly extracted funds from the exchange between the summer of 2011 and September 2013.

In a report published in April 2015 they wrote:

> Most or all of the missing bitcoins were stolen straight out of the MtGox hot wallet over time, beginning in late 2011. *As a result, MtGox was technically insolvent for years (knowingly or not), and was practically depleted of bitcoins by 2013.* A significant number of stolen bitcoins were deposited onto various exchanges, including MtGox itself, and probably sold for cash (which at the bitcoin prices of the day would have been substantially less than the hundreds of millions of dollars they were worth at the time of MtGox's collapse).[2]

In plain English, Mt. Gox had been running on empty long before it collapsed in 2014. It had enough capital to do business, but any slowdown

could mean disaster. The business it did, however, was substantial. By January 2014, Mt. Gox was handling millions of dollars in daily transactions, sometimes as much as $20 million in one working day.

If Karpelès knew just how precarious this situation was, he seems to have decided to plow on regardless. And neither the Financial Services Agency nor the Ministry of Finance nor the Bank of Japan did anything to regulate Mt. Gox, though all acknowledged that they knew of its activities.

It's a shame that Karpelès didn't read Spider-Man comics. As Uncle Ben said, "With great power there must also come—great responsibility!"

Karpelès had always loved Japanese manga but had relatively little interest in Western comics, so he may never have been exposed to this bit of comic book wisdom. For him, with great power came great wealth, and with great wealth, great silliness. He began to indulge himself, splurging on prostitutes, pet projects (of which, more later), and other amusements. He bought himself a custom-made bed that cost millions of yen. If he had learned to separate his personal account from that of his company and simply paid for stuff out of his own lavish salary, things might have gone better down the line.

The Mt. Gox site started accepting a dozen different currencies. Their first Japanese banking partner was Sumitomo Mitsui Banking Corporation (SMBC), but the man responsible for the account was reappointed elsewhere and his replacement felt uneasy about bitcoin. Mt. Gox switched to Mizuho Bank when SMBC was spooked by the unexplained decision taken by HSBC in Hong Kong to close Mt. Gox's account without returning its money.

More staff were hired to cope with expanding business, making it too expensive to stay at the Cerulean Tower. In the summer of 2012, the company moved to a different section of Shibuya, just behind the Cross Tower, on the fifth floor of a small building known as Round Cross Shibuya.

Twenty twelve was obviously busy. In order to tackle business operations in North America and to avoid the complex licensing regulations there, Mt. Gox signed a contract with a Seattle bitcoin service, CoinLab, in November, after a summer filled with negotiations. CoinLab was relatively new on the scene, but they had gained attention due to large funding from venture capitalists. They also managed the Bitcoin Foundation.

It seemed to be a good deal for both parties. CoinLab had a small but passionate team that could use the extra business and advice from an experienced company like Mt. Gox. And Mt. Gox, with CoinLab handling its US and Canadian clients, wouldn't have the hassle of getting a license to function there.

As it happens, four financial experts gave differing answers to the question of whether a license was required at all. Some said that bitcoin was not regulated, so a license would be unnecessary. Others said a license was required but would be nearly impossible to get. The money-transmitting-business (MTB) license covering all US states cost almost $50 million. Mt. Gox didn't have $50 million to invest at the time.

According to Karpelès, CoinLab assured him that they could handle the license situation.

The Financial Crimes Enforcement Network (FinCEN), an agency within the US Treasury Department, ruled in March 2013 that "a person is an exchanger and a money transmitter if the person accepts such decentralized convertible virtual currency from one person and transmits it to another person as part of the acceptance and transfer of currency, funds, or other value that substitutes for currency." FinCEN's mission is to safeguard the US financial system from misuse and to combat money laundering. In line with that mission, they took a hard line with money-transmitting businesses. The new "guidance" issued in March appeared to indicate that all money-transmitting businesses using virtual currency had to get an MTB license where anti–money laundering and know-your-client (KYC) measures were enforced and the people they did business with were identified.

A few months after the partnership began, problems started to arise.

On May 2, 2013, CoinLab filed a $75 million lawsuit against Mt. Gox, accusing it of not giving them full access to their North American clients, and continuing to serve customers there. Later that month, as part of the investigation into Silk Road, the US government seized a total of $5 million from Mt. Gox's accounts in North America and Karpelès's private account.

"With CoinLab threatening us legally," Karpelès said sadly, "Mizuho Bank recommended that we find another bank. Then a few weeks later, they refused to handle any outgoing international remittance transactions. We had no better alternative than to start using Japan Post Bank to do our computer transfers, at a maximum of ten per day, while our lawyers in the US were discussing with the prosecutors whether they could cancel the fund seizure."

Negotiations with other banks outside Japan did not work out. Eventually the company created a working relationship domestically with Japan Net Bank to solve their money-transfer issues.

To all appearances, Mt. Gox was still a success. It was still the largest bitcoin exchange. During the month of May 2013, it traded an average of $18 million a day, which was 70 percent of all bitcoin exchange transactions.

This wasn’t enough, though. Karpelès wanted more. He wanted to set up a bitcoin coffee shop to attract more Japanese users, with a staff of beautiful bitcoin baristas. He would call it the Bitcoin Café, with a proper accent above the *e*—he is a Frenchman, after all. People would pay for their hot coffee and croissants with bitcoins. It would show the Japanese people how simple bitcoins were to use. It would also act as a community center for bitcoin fans.

On August 29, 2013, Karpelès officially launched the café project. He even had the coffee mugs specially designed. But all this took more time than expected, and it wasn’t able to open on schedule.

Around November, he bought the company Shade 3D, allegedly in order to have a side business and guarantee the availability of cash in case he needed it.

And while he was busy with these sidelines, bitcoin was constantly in the news overseas.

On October 1, the FBI arrested Ross Ulbricht, the alleged founder of Silk Road. That same month, Mt. Gox acquired a money-service-operator license in Hong Kong. On October 29, the Las Vegas start-up Robocoin launched the world’s first bitcoin ATM.

In November, the bitcoin price exceeded $1,000 on Mt. Gox. In December, the number of verified customers on the exchange surpassed a million. The de facto value of a single bitcoin was whatever it was being sold for on Mt. Gox.

Mt. Gox had become the Dow Jones of the bitcoin world.

At this point, Karpelès had moved from his home in Tokyo’s Setagaya Ward to a new apartment in Meguro Ward due to domestic issues with his ex-wife, Kyoko. His new home was on the twenty-eighth floor of the luxurious apartment complex La Tour Aobadai.

Despite his wealth and growing fame, Karpelès was spending most of his free time eating junk food, watching anime, and compiling code. As his friend and colleague Julien Laglasse, a Frenchman living in Tokyo, says: “Mark is happy as long as he has a pizza, a coke, a computer to work on, and his two cats around him. He is not a greedy guy.” The trouble was, he wasn’t a careful guy either, and someone who was managing millions of dollars of clients’ money and a staff of over forty needed to be.

Before the shit hit the fan, someone on the Internet Relay Chat #mtgox publicly announced that Mt. Gox could be attacked via the method known as “transaction malleability” and explained how to do it. As a consequence of the warning, Karpelès improvised a way to block that attack immediately.

Later that month, he received an e-mail from a shady character offering to sell information about angry Silk Road vendors trying to attack Mt. Gox. He ignored it.

But there came a point when reality couldn't be ignored any longer. On February 7, 2014, Mt. Gox temporarily halted all withdrawals. The measure was taken due to the theft or disappearance of hundreds of thousands of bitcoins owned by Mt. Gox customers, as well as by Mt. Gox itself.

In a press release three days later, the company said it had suspended withdrawals because of a software flaw that would allow traders to defraud the exchange. However, what really happened was that Karpelès was beginning to confront the fact that a colossal sum was missing. To be precise, he was either just realizing that this had happened or found himself in a position where he had to admit the money wasn't there—and that it hadn't been there earlier. Only Mark Karpelès really knows the truth.

The announcement drew the ire of the bitcoin community because the flaw was allegedly well known and others in the business had already accounted for it. Mt. Gox was blaming the software when it should have taken direct responsibility itself. Its excuse reflected badly on the currency.

Jason Maurice, once at WizSec and now a freelance security adviser, believes that Karpelès misjudged the severity of the security issue and didn't implement a correct fix when it was needed. According to Maurice, it was only in early February 2014 that Karpelès understood the danger of the bug and came up with a proper solution, but by then it was too late. The damage had been done.

"Basically he dismissed a multimillion-dollar bug in his software that any decent software engineer would immediately have realized was a major issue," Maurice said in our talks with him. "Any other financial institution would have a quality assurance team to find something like that, but for Karpelès it was all up to him."

In addition to leaking money through the bug, the company might accidentally have been giving it away.

"Essentially, Mt. Gox was a dysfunctional organization," said someone who once worked for the company.

> Nobody was doing accounting reconciliation, and there was an exploitable fault in the transaction system that allowed people to get more or less paid twice. Think of it this way: if bitcoins were like frozen hamburger patties being served at a diner with a touchscreen menu, someone figured out a way of tapping the

> screen to get two for the price of one. Then one day somebody at the diner went to the freezer and realized they were completely out of hamburgers—and they'd only served half the customers they thought they had.
>
> The bitcoins were poorly secured, digitally and physically.

Former employees from the exchange claimed that at some point they stored about 90 percent of their bitcoins in paper wallets and USB keys. By moving bitcoins into a paper wallet—a printed document that contains all the necessary data to operate one or more private keys—the keys are no longer digitally stored where they might be subject to attacks. However, if the paper wallet is lost, the bitcoins in the wallet are also lost. Apparently, paper wallets at Mt. Gox were often haphazardly stored in the office, buried in sofas, or pushed behind desks. Karpelès denies this, saying that he shredded them once they had been inserted in the system "for security reasons."

According to another former employee, Mt. Gox "rented safety-deposit boxes in banks. When they needed to refill the transaction accounts, they took the bitcoins out of storage and deposited them in the system. There was no reconciliation in the accounting sense between the cold storage and the transactions done. As long as money was coming in at a steady pace, no one realized they might actually have been losing a lot of money. And when they did, all hell broke loose."

In February 2014, Karpelès informed this person that an estimated 850,000 BTC was unaccounted for—at the time, the equivalent of close to $462 million. He told him that users, exploiting flaws in the system, had probably siphoned off the bitcoins over several months. In particular, there seemed to be a system glitch that made it possible to get a payment reissued.

Teikoku Databank, Japan's largest and most respected credit-rating agency, had reviewed the company in July 2013, months before the collapse. The bank gave it a D4, the worst-possible rating a company can receive on their scale. One of the reasons for the low rating was the lack of qualified accounting staff at the company. There is also the possibility that Teikoku Databank simply had no idea of how to account for the value of bitcoins—a problem not unique to them.

Mt. Gox had survived hacks, system failures, and seizures by the authorities, but now things looked hopeless for them. Thousands of customers were unable to withdraw deposits, and Karpelès wasn't talking to the press. Speculation was rampant as to what exactly had happened, and the bitcoin world was in a panic.

Key members of the Mt. Gox staff and consultants gathered and

brainstormed for a way to keep the company solvent, protect its assets, and move forward. They drafted a document—a “Crisis Strategy Draft”—that was meant to show investors the problems and possible solutions. On February 24, Karpelès resigned from the board of the Bitcoin Foundation, of which he was a founding member along with Gavin Andresen (the chief scientist), Charlie Shrem, and Roger Ver, among others. He reportedly told the organization of the troubles on the horizon.

The final nail in the coffin was the unauthorized release of the Crisis Strategy Draft, the supposed plan for saving the company. The document was unfinished.

“The Crisis Strategy Draft had only been shown to a few people, including the Winklevoss twins, who were active investors in the industry and SecondMarket executives. If, prematurely, it got into the public domain, it would be disastrous,” another former employee said. A few days after being put together, the document was leaked to the blogger the Two-Bit Idiot, who published it on the web on Monday, February 24, 2014, at 6:23 p.m., EST. It spread across the Internet within hours.

The media were soon all over the story. (It was at this point that we began covering it.)

If the document hadn’t been leaked, Mt. Gox might have survived.

“Our last-minute efforts to discreetly refinance the company and avoid insolvency had pretty much been sabotaged by it,” the same employee said.

Mt. Gox suspended all trading after internal investigations discovered a loss amounting to 744,408 BTC. The accounting practices at the firm were so slipshod, however, that even what seemed like an “exact” sum wasn’t accurate. On February 25, after confirming with their lawyers that the end was near, they shut down entirely. Since Karpelès still wasn’t sure exactly how many bitcoins were missing, he started scanning the paper wallets and confirmed that the cupboard was bare.

On February 28 at 11:39 a.m., EST, the *Daily Beast* published “Inside Japan’s Bitcoin Heist,” reporting the amount of missing bitcoins to be 820,000 BTC—the closest figure yet to Karpelès’s initial calculation.

Mt. Gox filed for bankruptcy protection with the Tokyo District Court. It declared liabilities of about ¥6.5 billion ($64 million at the time). The company said they had lost almost 750,000 BTC belonging to its customers and about 100,000 BTC of its own, then worth around $462 million.

Karpelès remembers that day well. “I went to the Tokyo District Court with my lawyers in the afternoon and, by the end of the afternoon, the order

was given. With just thirty minutes to prepare, we then gave a press conference. The room was full."

He answered the questions in his French-accented Japanese without forgetting to make the deep bow of atonement and apology expected of a CEO.

"The press conference in itself went okay, although one cameraman decided to ask questions in English, which I wasn't prepared for. But the worst thing was the walk to the taxi outside, with all the reporters surrounding me, getting closer and closer. Once in the taxi, the driver had to struggle hard to get out of there. I was followed all the way to the Baker & McKenzie offices, where I was finally able to rest a bit before driving home."

In its official statement, Mt. Gox said that the bankruptcy was related to a bug in the bitcoin software algorithm that was exploited by one or more persons. "We believe that there is a high probability that these bitcoins were stolen," it asserted bluntly, blaming hackers. And since the company had a business plan that looked forward to 2017, it seemed unlikely, some felt, that its CEO had stolen his clients' money himself.

After filing for bankruptcy, Karpelès went on working at the same office in Shibuya. The number of employees was drastically reduced. There was a small coffee shop on the first floor that had been destined to be the world's first Bitcoin Café. Today, it is still just a café.

One of the former employees who believe that Karpelès did not act maliciously or for his own profit says: "He's a workaholic and a geek, but a good-hearted geek. He just has very limited management skills, a little hubris, and didn't pay attention to accounting. He was only twenty-seven or twenty-eight years old."

A colleague was more complimentary. "He wrote most of the code—he created a fantastic application interface. It's a wonderful platform for trading bitcoin. The problem isn't bitcoin— it's the way Mt. Gox was run. And it was run into the ground."

At the time of the bankruptcy, Yoshihide Suga, a top government spokesman, stated that official organizations including the police and Japan's Financial Services Agency were collecting information on the bitcoin trade in Japan and considering regulatory action. In addition, there were reports that US authorities had begun investigating Mt. Gox and had subpoenaed individuals who had worked for, or still work for, the company.[3]

The United States was interested in Mt. Gox for many reasons. The Silk Road investigation was ongoing, and some law enforcement agencies

believed that Karpelès might have been the victim of a group of cybercriminals they were looking for.

A few days after the bankruptcy, we spoke with Karl-Friedrich Lenz, professor of German and European law at Aoyama Gakuin University in Tokyo and author of an academic paper entitled *Legal Issues of the New Internet Currency Bitcoin in EU Law and German Law*. Professor Lenz believes Mt. Gox should have been treated as a banking institution and not allowed to operate without a license under Japanese law at the time. He believes that because Mt. Gox accepted deposits and money could be wired to its accounts, it was more or less a financial institution. “I went to the Financial Services Agency and asked to speak with someone on this issue. After much argument, someone from the banking division accepted my report, which I hope they will review.”

He added, “If Mt. Gox had been treated like a bank, this problem would have never happened. It would have had to have proper accounting and people with financial skills to get licensed. It is unlikely that the Japanese government would have granted such authority to a company run by a twenty-seven-year-old geek with no background in finance.”

Karpelès recalls that, in the days following the bankruptcy announcement, things calmed down while the trustee appointed by the Tokyo District Court was figuring out what to do.

“We had to rapidly put in place a call center, which I developed within a couple of hours, the best I could, and then things started to move along.”

The trouble was, any hope of civil rehabilitation of the company was hampered by Karpelès’s inability to travel out of Japan, since he had to be available at any time to answer the liquidator’s questions.

In the United States, his lawyers filed Chapter 15, which is done when a liquidation or rehabilitation process is taking place in another country. As part of routine procedure, the US courts needed to question Karpelès in person; otherwise, the filing would be impossible and the assets of the company in the United States would be threatened with seizure.

A rumor went around that Mark was planning to travel to America, which prompted FinCEN to submit a subpoena and the DHS to dispatch a couple of agents to each airport he might use. He had become a person of interest. Of course, it was out of the question for him to travel under these conditions. And the court in Tokyo wouldn’t allow him to leave the country without a guarantee that the United States would permit him to return. So Mt. Gox wasn’t authorized to proceed to rehabilitation, and liquidation began around the end of April 2014.

“After that, I tried to make Tibanne, my first company, work, but without great success,” Mark said.

Karpelès noted that while the general public knew about most of the big events connected with Mt. Gox’s downfall, many details went unnoticed. “All I can say is that each and every move taken by Mt. Gox and Tibanne was done for a valid reason.”

On March 20, 2014, the discovery of an old wallet brought the number of bitcoins still missing to roughly 650,000, down from about 850,000. A statement posted by Karpelès online said: “MtGox Co., Ltd. had certain old-format wallets which were used in the past and which, MtGox thought, no longer held any bitcoins.” This was both good news and bad. People wondered: If Mt. Gox could simply misplace 200,000 BTC, maybe they had misplaced a whole lot more. Maybe Karpelès was hiding them.

Many mysteries remain. Independent investigators are focusing on the way Mt. Gox was bought by Karpelès from McCaleb. Some believe that the theft of the bitcoins occurred during the period when transfers to paper wallets were being made. If so, it happened precisely when detecting any theft would be the most difficult.

After the bankruptcy, Karpelès’s lawyers filed a report with the Japanese police requesting an investigation into the hacking of the company and the missing bitcoins. Not everyone was confident, though, that the police had the skills to solve the case, and some Mt. Gox creditors launched their own probe. This was something Karpelès welcomed.

> I totally approve of there being several separate investigations. It’s generally a good idea to have different people looking into the same problem from different angles. The police don’t report in detail what they are doing, so it can seem that they aren’t doing anything. But they are actually making progress. What’s worrying, however, is recurrent evidence that innocent people have been arrested and made to confess to crimes that they didn’t commit. So I just hope they won’t do anything crazy here. There’s no guarantee, of course.
>
> Personally, I support their efforts to find the culprit or culprits. The method used is less important to me than the result.

Unfortunately for Karpelès, the result wouldn’t be quite what he expected.

CHAPTER FIVE

THE UNUSUAL SUSPECTS

After the shutdown of Mt. Gox, the bitcoin community and the media vilified Mark Karpelès as a clown, a con man, and a horndog. Japanese tabloid magazines portrayed him as a "beast" hiding in his "lair" and tracked him to his new address, a modest house in Toshima Ward, where he'd moved when money started to get tight.

His reputation was damaged further when, as 2015 started, the Silk Road trial began.

During the trial in a Manhattan federal courtroom, Ross Ulbricht's lawyer immediately tried to peg Karpelès as the real Dread Pirate Roberts. Karpelès denied the accusation and said he had cooperated with US federal law enforcement.

"This is probably going to be disappointing for you, but I am not Dread Pirate Roberts," he said.

> The investigation reached that conclusion already—this is why I am not the one sitting [on] trial, and I can only feel defense attorney Joshua Dratel trying everything he can to point the attention away from his client.
>
> I have nothing to do with Silk Road and do not condone what has been happening there. I believe bitcoin [and its underlying technology] is not meant to help people evade the law, but to improve everyone's way of life by offering [never-thought-of-before] possibilities.

DHS Special Agent Jared Der-Yeghiayan, under cross-examination, said he had pursued Karpelès as the suspected owner and operator of Silk Road in 2012 and 2013. But Der-Yeghiayan acknowledged that the things Karpelès may have written online did not match the level of English skills that Dread Pirate Roberts possessed. He thought it was someone else close to him, a person who was working for him by the name of Ashley Barr. There is no solid proof behind this assumption, but if there had been an award for disgruntled employee of the month at Mt. Gox, Barr would probably have won it twelve months out of the year. There was certainly bad blood between him and his boss.

Other sources close to the DHS told us that Karpelès had been eliminated as a suspect in the running of Silk Road by the end of 2013 and

confirmed that he had been generally cooperative with the government. He had notified them about discovering over a hundred fake American passports being used to set up Mt. Gox accounts and debarred the clients. He also acknowledged owning a web-hosting service used by part of the Silk Road network.

Dread Pirate Roberts, who seems to have known Karpelès was cooperating with the authorities, had tried to reach Karpelès many times in the months after Silk Road was closed down. On September 1, 2013, one month before Ulbricht's arrest, he sent several e-mails to Karpelès's bitcoinfoundation.org address with the header "info regarding active law enforcement investigation." One read:

> Greetings Mark
> As the subject says, I have some information for you as well as a special request. I don't want to say any more without PGP protection. Do you have a public key I could use?

Dread Pirate Roberts also tried to make contact with Mt. Gox employee Ashley Barr. Karpelès didn't respond, not being in the habit of using that e-mail link. But this didn't stop the man from trying to contact him, presumably because he urgently needed to know how much the authorities knew about his operation.

Someone had tipped him off that Karpelès was working with the good guys. In the uncensored version of Ulbricht's diary, popularly dubbed *The Dread Pirate Journal*, Ulbricht wrote that an entity he called "French Maid" claimed Karpelès had given the DHS information about Dread Pirate Roberts. An entry for September 13, 2013, said Roberts was offering $100,000 for information about the joint investigation by the federal authorities into Silk Road.[1] A day later, French Maid was paid this sum by Dread Pirate Roberts.

Though an edited version of the diary was submitted as evidence at the trial, the prosecution had removed almost all information about Dread Pirate Roberts buying information from what may have been law enforcement sources. It was obvious they were unhappy about the existence of French Maid. But while they censored the pdfs, they forgot to redact embedded information in other files, making it possible to read *The Dread Pirate Journal* without official interference. Journalists and investigators rejoiced. There was enough there to allow people to figure out how to access Dread Pirate Roberts's many e-mail accounts and see the evidence for themselves.

So, who was French Maid? Karpelès had some idea. It was only after Ulbricht's conviction that the world would find out.

On March 30, 2015, the Department of Justice released documents about

two federal agents with key roles in the pursuit of Silk Road who allegedly stole proceeds from the underground site in the form of bitcoin. They hid the illicit earnings while continuing their work.

US law enforcement sources confirmed that one of the agents, in exchange for a large payment in bitcoin, also apparently leaked information to the Silk Road operators on the extent to which Karpelès was cooperating with the Silk Road investigation. In a later e-mail, Karpelès wrote: “The fact that information provided by our lawyers to law enforcement ended up being known by parties being investigated [before arrest] is a very big issue.”

The agents, in addition, made an attempt to extort funds from Mt. Gox. They demanded that Mt. Gox do business with them, but they were refused. In response, and at their urging, the US government seized a total of $5 million from the company’s US accounts. The seizure seemed to be more motivated by personal interest and the rogue agents’ wanting to cover their tracks than by the pursuit of justice.

These incidents raised questions about the events leading to the collapse of Mt. Gox, and they are likely to put a chill on those who might wish to cooperate with government investigations in the future.

According to court documents and government sources, former Drug Enforcement Administration Special Agent Carl Mark Force IV, who was arrested on March 30, 2015, asked Ross Ulbricht to pay him $250,000 in bitcoin not to disclose information to the government. Force also sold Ulbricht information about Karpelès. US government sources told us that they believe the rogue agents were the first to inform Silk Road administrators that Karpelès was lining up with the investigation and had shared information on potentially illegal activities with federal authorities in the past.

US federal court documents outlining the charges against the two agents also assert that former Secret Service employee Shaun Bridges stole more than $800,000 in bitcoins from Mt. Gox during the investigation into Silk Road.[2] Supposedly, he wired the funds from a Mt. Gox account he controlled to a personal investment account over the course of nine transfers from March 6 to May 7, 2013, days before Mt. Gox’s US funds were seized—an operation in which he played a major role. Bridges resigned on March 18, 2015. He pleaded guilty in federal court in August of 2015 and was later convicted. In his plea agreement, he also acknowledged that he obstructed justice by “creating an additional incentive for Ulbricht to attempt to hire someone to kill a cooperator [with the investigation] whom Ulbricht suspected of committing thefts I had in fact committed.”[3]

The US Department of Justice released a statement on August 31, 2015,

in which Richard Weber, chief of IRS–Criminal Investigation, put it bluntly, “Through a series of complex transactions, the defendant [Bridges] stole bitcoins worth hundreds of thousands of dollars . . . This case is an excellent example of the financial expertise of our special agents. Through the analysis of both the block chain and data from the Silk Road servers, we were able to trace the flow of funds, which eventually led to the defendant.”

Even the most amateur sleuth would probably see this as an opportunity for Bridges to cover his tracks. With the US accounts under the control of the investigators, it is believed that he hoped to gain access to and possibly erase any record of his transactions with Mt. Gox. But there may have been another reason the agents orchestrated the seizure of Mt. Gox funds: sour grapes.

As early as June 2011, Karpelès had volunteered information to the DEA. In a letter received by them on June 13, he wrote:

> We are a Japanese securities exchange known as Mt. Gox. [We are] contacting you in light of the recent media excitement surrounding Bitcoin.
>
> It has been brought to our immediate attention that there are U.S. senators calling for quick and firm action from the DEA to shutdown illegal narcotic and international drug trafficking services that are facilitated by a website known as the “Silk Road.”
>
> Firstly, we would like you to be aware that we do not condone the use or trafficking of illegal substances, nor does our bitcoin exchange directly or indirectly endorse the use of such services. Ultimately, we are pursuing a goal of accepted legitimacy, both for bitcoin and our exchange. We are more than willing to comply with any court-sanctioned investigations regarding our services.
>
> We invite you to contact us.

While Karpelès seemed to be doing his best to stick to the straight and narrow, however, two special agents working the Silk Road case were more concerned about exploiting the situation than indulging a well-meaning Frenchman.

E-mails provided by Karpelès and also reported in *Forbes* suggest that Force attempted to establish a business partnership with Mt. Gox. Federal law enforcement sources said they checked into them and believe they are authentic.

According to the e-mails, on April 8, 2013, Force attempted to make contact with Karpelès through a LinkedIn invitation. The message he sent to

him even mentioned that he was a DEA special agent. After being ignored, Force apparently tried again on multiple occasions to do business with Karpelès. (He resigned from the DEA in May 2014.)

On April 11, 2013, Force wrote:

I don't think we connected on LinkedIn?

I wanted to ask you if you could . . . back me on a deal for 250 Bitcoin. A sale.

Best regards,

Carl "Mark" Force

On May 7, he allegedly offered to become the representative of Mt. Gox in North America. His e-mail said:

I saw the news yesterday that you won't be partnering with Coinlab.

Sorry to hear that.

If you are still looking for a U.S. and Canada representative, please keep me in mind.

Thank you very much,

Carl Force

In mid-May, the DEA seized a total of $5 million worth of Mt. Gox accounts in North America without giving any explanation to the company.[4]

On May 15, 2013, as if to make him regret not considering his offer, Force wrote to Karpelès: "Told you should have partnered with me!"

Cynics would say, "No good deed goes unpunished." In the case of Mark Karpelès, this seems true. His apparent willingness to play ball was very dangerous.

According to court documents and reports, Ulbricht had zero tolerance for interference with his business. He was vengeful enough to have allegedly paid the equivalent of $80,000 to the rogue federal agents, without knowing that they were government officials, to arrange the murder of a person he believed had betrayed Silk Road and was working with the authorities.[5] Ulbricht, however, was not convicted on murder-for-hire charges.

Revenge may also have been behind the larceny at Mt. Gox. A cybersecurity consultant working with the Tokyo Metropolitan Police Department said, "In light of the evidence that federal agents made the Silk Road operators aware of cooperation with the investigation by Mt. Gox staff,

we should have considered the possibility that the bitcoin theft at Mt. Gox was an act of retaliation, possibly with help from an insider."

During the Silk Road trial there were repeated attempts to identify Karpelès as the Silk Road founder, which he lightly dismissed, saying, "I am not a pirate or a fraudster." But a new accusation that he couldn't shrug off was soon leveled at him.

On January 16, 2015, bitcoin millionaire and Belgian tech entrepreneur Olivier Janssens publicly accused Karpelès of making death threats against one of his former employees when Mt. Gox was still in business.

Janssens, a former client of Mt. Gox, gained fame a year earlier for becoming the first person to pay for a flight using bitcoin. He made some of his fortune through Mt. Gox at a time when the company was flourishing. Unfortunately, he claimed to have lost the equivalent of $5 million when Mt. Gox filed for bankruptcy.

In a tweet, Janssens wrote: "I have black on white proof of Mark Karpeles making multiple death threats against an ex-employee. So [do] Japanese police. #RossUlbricht."[6] The hashtag was a reference to the fact that Ulbricht was also being accused of plotting to kill a former employee. The accusation sparked a torrent of comments on social media.

It was likely that Janssens planned to give the "evidence" to Ulbricht's lawyer. Luckily for Karpelès, the judge in Ulbricht's case, Katherine B. Forrest, ruled the claim of Karpelès's involvement with Silk Road as speculative. Nor would she allow future testimony regarding the possibility of his being Dread Pirate Roberts. In addition, she struck previous testimony about Karpelès being a suspect from the record. Though this was obviously good news for him, the mere mention of Karpelès in this light can't have done his reputation any good.

The evidence Janssens planned to bring against him included e-mails allegedly sent by Karpelès to a former Mt. Gox employee. One, dated Saturday, March 1, 2014, read:

> Keep in mind the warning letter you received. If you [make any] statements that contain confidential information, we will find a way to deal with it, maybe outside of the law, by making you return home as a dead man, for example. I'd appreciate it if you keep quiet.
>
> Thanks.

This was supposedly signed by Karpelès and sent from his e-mail address. A second threatening e-mail was apparently sent a couple of days

later, on Monday, March 3, 2014, which read: “We are watching you closely. Always remember that.”

It wasn’t signed.

Intimidation (*kyohaku*) and extortion (*kyokatsu*), depending on the nature of the threat, are criminal offenses in Japan.

However, in an odd turn of events, Janssens then retracted his accusation and announced that his allegations were being followed up by people on his own team. A cybersecurity expert close to Janssens admitted that the e-mails could have been forged by the recipient or even a third party. Janssens’s lawyer, Takaaki Nagashima, said he could not comment on the accusation because the complaints were made by a former Mt. Gox employee, who, like all the other staff there, was a suspect in what might have been “an inside job.”

Karpelès denied ever making death threats against his employees. He claimed to be confident that forensic analysis of the e-mails would show they were forged. The e-mail system at Mt. Gox used Google, and all outgoing e-mails were signed through DomainKeys Identified Mail, a validation system designed to recognize fake e-mail messages. Ideally, researchers could study this facet of e-mail technology to see whether Karpelès did actually write these messages.

In a written statement of January 20, 2015, Karpelès’s lawyer in Tokyo asserted that the accusation had no basis in fact: “There are individuals circulating amateurish forged documents as if they were real evidence and we cannot overlook this.”

Japanese police sources said they were aware of the charge against Karpelès, but declined further comment at the time, saying the investigation was ongoing.

The question remains: Why would an ex-employee falsely make this claim?

Threats of another kind had been made against Karpelès himself.

On February 26, 2014, one day before Mt. Gox went completely blank, an attempt to extort money from him was made by “CARPENOCTEM,” warning him that “angry Silk Road [1.0] vendors were acting against Mt. Gox.” The e-mail message reads in part:

> Date: Wed, 26 Feb 2014 12:04:45 +0000
>
> You are writing/dealing with a criminal. And btw: I know you are not a criminal. You bailed out other sites in the past, and I am sorry

you get bashed that much . . . Feel free to drop me a bitmessage:

CARPENOCTEM was right about one thing: Silk Road's vendors, and even its operators were furious with Karpelès. And the situation for him was only going to get much worse.

CHAPTER SIX

THE OTHER UNUSUAL SUSPECTS

At the end of 2014, the Cybercrime Unit of the Tokyo Metropolitan Police Department leaked to the press that the collapse of Mt. Gox was due to an insider. The *Yomiuri Shimbun* ran it on the front page on January 1, 2015. But who was the insider? Was there an accomplice?

Data leaked and available after Mt. Gox's shutdown was analyzed by a group of independent investigators from WizSec, led by Kim Nilsson, a Swedish software engineer, and Jason Maurice, both based in Tokyo. Their interim report released on February 14, 2015, gave the clearest account yet of what could have happened to Mt. Gox. The report asserted that the Mt. Gox hack was mainly an inside job.[1]

It also detailed the activities of a robot—that is, a program embedded in the Mt. Gox system—that had been buying hundreds of thousands of coins with fake money within the company. Robots or "bots" probably account for a significant amount of the missing bitcoins.

Days later, maybe in response to this report, a hacker or group of hackers anonymously leaked more data, including private information about Mt. Gox creditors, that should only have been in the possession of official investigators and the trustee who was liquidating Mt. Gox. At the same time, the alleged hacker(s) attempted to extort money from users, saying there was more information that could be released if they were paid. The message was immediately taken down, possibly by the moderators of the site where it appeared—Reddit, a popular Internet bulletin board.

Among the new information the report provided was an analysis of the time frame involving "Willy bot," a program in the Mt. Gox system that seems to have started sometime in 2013 and was active for bouts of twenty-one hours a day—except between 2:00 and 5:00 a.m. Japan time (12:00 to 3:00 p.m., EST).

The purpose of Willy bot is still not completely known. Did it exist to cover up losses or steal bitcoins? And who put it there?

Maurice asserts that the activity was clearly intentional and illegal. The bitcoin trading data showed that automated software, with database access, was used to create new accounts at Mt. Gox with fake money, the balance being set at millions of dollars. The bot then spent that fake money on the exchange to buy up a few bitcoins every few minutes, usually over the course of a day or so. Spending $2 million to buy bitcoins off the market is

something that had to have been done slowly, Maurice postulates.

There is no evidence that points to the hacker's nationality. Maurice asserts that there is strong evidence that the trading bot was being run from either inside Japan or within the Mt. Gox network itself.

The times of the robot's operations also point to an inside job. This suggests that there could have been more than one person behind the bot, working in shifts.

Kristian is a software engineer who was a PhD student at Tokyo University of Technology at the time he was trading bitcoins on Mt. Gox. He was the first to investigate Willy bot. He said the bot was known to traders quite early on and was dubbed "Willy" on the regular discussion forum BitcoinTalk after the killer whale in the movie *Free Willy*.

Kristian first noticed it in November 2013. The robot was steadily buying about ten to twenty bitcoins every five to ten minutes. When we interviewed him, he told us, "At the time, I didn't think it [had been] placed there from [inside] Mt. Gox. There was nothing that indicated that there might be something shady going on, although it was suspicious that someone would buy that much. Because it was at least clear that it was the same entity constantly buying."

Kristian's curiosity was aroused. "I was trading [myself and] was really into it, and at some point it became really noticeable. I wasn't the first person to notice it. There was something out there that was buying all the time at a predictable pace. I assumed that was why the price was going up."

He started investigating the matter more closely two months after the Mt. Gox trading data was leaked on the Internet, in March 2014, and he published a blog post about Willy in May of that year. It became apparent to him that Willy was perhaps located somewhere in a Mt. Gox server, because even when the company's API (application programming interface) was down and nobody could trade, the bot kept going.

Based on Nilsson's research, it seemed possible that the administrators of Mt. Gox, after losing large amounts of bitcoin, were running the bot themselves to make up the shortfall. "My guess at the time was that the bot was used to manipulate the market to bring up the price" and raise the value of "whatever Mt. Gox was holding." But later research made him fairly certain that Mt. Gox had not accumulated a bitcoin stockpile.

Kristian analyzed what parts of the Mt. Gox trading system were involved with bots. This led him to link the Markus account—an account that, along with Willy, controlled the fail-safe, money-replenishing system set up

by McCaleb to counter breaches of the exchange—with Karpelès's private account. And he noticed that the Markus account stopped trading immediately before the Willy account started trading. Consequently, it seemed very likely that Willy was operated by the same person who was operating Markus. Since Markus was believed to be Karpelès's administrator account, conceivably someone had appropriated it and put it to different use. Neither Kristian nor Nilsson could say with any certainty that it was actually Karpelès controlling it.

Karpelès himself says he was unaware of this activity.

The WizSec report shows that there may have been multiple hacks. One of the attacks could have lasted several years and manipulated the bitcoin market. Another was used simply to steal the coins.

"Until we get the full database, we can't really be sure," Maurice said. "There could have been multiple hackers in multiple countries. It could have all been [done] inside Japan and we don't know. But there is a lot of evidence pointing to inside Japan."

Meanwhile, the Japanese police were going their own way about it.

There have always been rivalries in the Tokyo Metropolitan Police Department, and at some point in 2014, the White Collar Crimes Department (捜査二課, *Sosanika*) took the lead on the Mt. Gox case away from the Cybercrime Unit. Perhaps the top brass felt the case was moving too slowly or that whatever had been learned so far wouldn't satisfy public opinion. Sources say the White Collar Crimes Department decided to go after Karpelès on counts that could be easily proved, and his sloppy bookkeeping opened him to charges of embezzlement and improper use of electronic transfers.

In Japanese police lingo, they were going to apply *bekken taiho* (arrest on a separate charge). The methodology is sound: arrest a suspect on some minor charge and, while you have them in custody, interrogate them at leisure about the major crime you suspect them of committing.

This is why in Japanese homicide cases the suspect is often first arrested for improper disposal of a body, then, after weeks of interrogation, arrested again on the charge of murder.

In early June 2015, Japanese journalists, hearing rumors of an imminent arrest, started to gather openly in front of Karpelès's residence, stalking the people going in and out of it. By the end of the month, a Japanese reporter from the *Nikkei* newspaper informed Karpelès's lawyer that the police were planning to arrest him. However, for two months, nothing happened. At some point, Karpelès himself decided to call up the cops in the Cybercrime Unit,

with whom he was in touch anyway. “I hear rumors that I’m going to be arrested soon,” he told them. “If this is the case, please let me know when, and I’ll drive over to the police station. I’d like to avoid a scandal in this quiet neighborhood.”

The detective he spoke to replied that no such thing was being considered, and that Karpelès should relax. “No one is going to arrest you, and if that was the case, I would be the first to be informed.” This was about twelve hours before the actual arrest. The police evidently wanted to be sure it was done in as public a fashion as possible.

Karpelès was pretty confident that the rumors were false—after all, they had been wrong before, and he’d had a meeting with the Japanese police, Interpol, and the FBI only a few weeks before. It was a Friday, and he was preparing some homemade lasagna for some friends. Around midnight, he went to bed, according to people he’d been chatting with online. Less than six hours later, on August 1, 2015, the cops knocked at his door, and the house was surrounded by about a hundred journalists and police officers.

Around six in the morning, Karpelès was paraded before the throng of reporters. He seemed calm. He was wearing a light-blue T-shirt that had “Effortless French” written across it and a hat, black on one side and white on the other.

The T-shirt had been a gift from a friend who sensed that things weren’t going to work out well. The friend had told Mark to *ganbare*—the all-purpose Japanese word that means everything from “good luck” to “work hard” to “tough it out.”

When he left the house, he kept his mouth shut. His curly dark hair was tied in a ponytail, a new look he had adopted that year. His hands, cuffed, were hidden under a white towel with the words “Don’t Panic” on it. This was from his favorite book, *The Hitchhiker’s Guide to the Galaxy*. His cats were left alone inside the house for about twenty-four hours before someone was allowed in to feed them.

The arrest charges were amazingly indirect. In a statement, the police said they believed that Karpelès had illegally manipulated transaction records on a system that Mt. Gox used to swap bitcoins for dollars and falsely transferred $1 million into “a certain account.”

The police did not clarify where the $1 million had come from or what Karpelès had intended to do with it.

The *New York Times* pointed out the bizarreness of the initial charges in their coverage, noting, “But the arrest, and the small amount of information

divulged by Japanese law enforcement officials, shed little light on the larger mystery of the missing Bitcoins . . . In any case, the amount cited by the police represents only a tiny fraction of the value of the Bitcoins that went missing from Mt. Gox."[2]

Everyone was surprised by the arrest.

And the charges.

Even Karpelès's lawyer.

No one was prepared.

Because I (Jake Adelstein) knew Karpelès's lawyer, Nobuyasu Ogata, from a previous story I had done relating to a different client of his, I ended up, for a short time, with the job of feeding Karpelès's cats. This wasn't easy for someone with a severe cat allergy. Nor did I enjoy wading through the army of Japanese reporters parked in front of Karpelès's house who were surprised and horrified to see a fellow journalist entering and leaving the suspect's home. In some effort at transparency, I tweeted pictures of myself with the cats.

The public may not have been fond of Karpelès, but his cats seem to have acquired a solid fan base. The two felines, Tibane and Julia, were eventually adopted by a friend and colleague of his, Julien Laglasse.

His lawyer requested bail, pointing out that Karpelès had cooperated with the police and provided them with evidence. The prosecutors asked for bail to be denied, claiming that Karpelès would destroy evidence or cut and run, or do both. As is often the case, the judge ruled in favor of the prosecution.

He remained in the holding cell. He was not allowed visitors. He was only allowed to read books that passed inspection. His time with his lawyer was limited.

On September 11, 2015, the prosecutors finally filed the first criminal charges against Mark Karpelès. The alleged crimes were embezzling over $2.7 million of his clients' money and illegal manipulation of electronic data.[3] The police had detained him, at that point, for more than forty days.

One Tokyo-based accountant said that if the prosecutors were charging him with embezzling this amount of money, the majority of other businesses in Japan could be charged for the same crime.

Ironically, the cell in which he was held was in a police station located in Akihabara, the electronics mecca where he had spent so many happy hours. While kept in detention, he lost thirty-five kilos (seventy-seven pounds).

In Japan, the police can keep arresting you on separate charges for periods of up to three weeks each. The longer you hold off from telling them what they want, the more charges they throw at you, and the longer you can be interrogated. During the interrogations, a lawyer is not allowed to be present.

Karpelès's interrogation had some bizarre moments. According to those close to the investigation, the interrogators kept asking him over several days if he was actually Satoshi Nakamoto. Karpelès did not confess to this.

In the many charges against him, the prosecutors did not even address the missing 650,000 BTC (worth an estimated $460 million at the time they were lost). He may have been a sloppy manager, but that in itself is not a crime. In other words, nothing in the charges filed pertains to the missing bitcoins or sheds any light on how the company collapsed because of the loss.

At the end of October, the police arrested him again on embezzlement charges. The judge verbally reprimanded the prosecutors, telling them to stop dragging out the case.

Eventually, he was also accused of moving ¥20 million ($166,000) in client money to his own bank.

Karpelès's legal team at the Ogata Law Office in Tokyo said that the accusation of embezzlement was unfounded, claiming that the money came from company income and his salary, to which he was entitled a total of ¥341 million ($2.8 million). The act he was accused of, they said, did not even meet the technical specifications for embezzlement.

While his defense team was preparing for the court case, the police were busy making their case solid in the court of public opinion. Based on police leaks, the *Yomiuri Shimbun* reported on October 28, 2015, that the defunct firm's chief had spent an unspecified sum on prostitutes, as did other newspapers, citing the police, who said this involved "several women whom he met at venues that offer sexual services."[4]

Karpelès, a Japanophile, may have been expected to behave like a good Japanese suspect and confess to the charges against him—even if he hadn't committed any crime—so they could close the case. That's what innocent people had done in another big cybercrime case. That case, ironically, had also involved a portly computer geek with a fondness for cats.

The difference here was that the geek might actually be innocent.

CHAPTER SEVEN

PRESUMED GUILTY UNTIL PROVEN GUILTY

Ask yourself this: If Mark Karpelès had 650,000 BTC stored away somewhere, why hadn't he run off to a desert island like Roger Ver? He could have been living it up in South America or somewhere with a new passport and a new life instead of rotting in a Japanese holding cell. If he was guilty, why did he work so closely with the Japanese police and give them access to all his data? Why not erase the data before going bankrupt and get away with being charged with a relatively minor offense?

This strategy had worked brilliantly for the head of Incubator Bank of Japan several years before.

Incubator Bank of Japan filed for bankruptcy in September 2010 at the Tokyo District Court after overexpanding and engaging in a number of allegedly illegal transactions, which resulted in it being closed down. The collapse could have rocked the foundations of Japan's finance system, but the media downplayed it. It was the country's first bank failure in seven years, and also the first case since the government limited the deposit guarantee to ¥10 million. Incubator Bank had excessive liabilities of ¥180.4 billion.

Takeshi Kimura, the chairman of the bank, ordered records to be destroyed as he got wind of the Financial Services Agency (FSA) getting ready to launch an investigation. He was found guilty of obstructing an FSA inspection and given a suspended one-year prison term. He never even served time in jail. It was a case that proved that if you're well connected and eliminate evidence, you may get away with being charged for a much lesser crime.

Karpelès could have tried a similar tactic. He could easily have wiped out almost all the transaction records and all the data in the servers, leaving little to investigate except for banking records. He didn't. He gave the data to cybercrime experts and the police, and he personally tried to figure out what had happened.

Kim Nilsson, the cybersecurity pro who has now spent several years looking at the Mt. Gox problem, isn't sure who created Willy, the trading bot that may have generated huge losses for Mt. Gox, but he is sure that there was a massive theft of bitcoins—perhaps several thefts—which the Japanese police have failed to fully investigate.

He agreed to answer our questions for the book, but on condition that we vetted the results with him, to make sure there were no misunderstandings and

because he was still involved in the case.

> This is the discussion that took place:
>
> Q: Do you have an idea of the scale of the theft?
>
> A: It's counted in the hundreds of thousands of bitcoins. It started in autumn 2011, and continued throughout 2011 and 2012. In 2013 it slowed down, mainly because Mt. Gox was empty of bitcoins.
>
> Q: Did the theft occur through a bot or hacking?
>
> A: We can't be one hundred percent certain, but probably someone hacked Mt. Gox and in one way or another got access to the private keys.
>
> Q: Where the hacking comes from is very difficult to figure out in this day and age. Do you have any idea where it was from? Is it even possible to say geographically?
>
> A: Many people online like to speculate that maybe Mark Karpelès took all the bitcoins himself, but I doubt he would have carefully siphoned off money in this fashion when he could have just spent it directly. Mt. Gox was compromised by hackers on several occasions [as shown by the leaks of internal data], so it's to be expected that hackers were behind the running theft as well. As for their location, that's just one of many things that has to be deduced indirectly from the available evidence.
>
> Q: When will you be able to definitely find out where the bitcoins went?
>
> A: Figuring that out, in terms of the blockchain, isn't very hard. Let's imagine that a thousand bitcoin addresses were used to receive the stolen bitcoins. Identifying the thousand addresses is the relatively easy part [given that you have enough data]. The tricky bit is identifying who *owns* the addresses.

Nilsson has gigabytes of data and documents that show Mt. Gox was hacked and bitcoins were stolen. But for the sake of argument, let's suppose it was all a ruse—that Karpelès was pretending to seek the real culprit to cover his own guilt. To refute this, a special agent working with Homeland Security told us in August 2016:

> I believe that the majority of the hacks took place while Mark Karpelès was moving the bitcoins into cold storage. It's an ongoing investigation that involves US jurisdiction so I'm limited in what I can say, but there was a flaw in the process and in the system he

> used to store the bitcoins. Now it could be argued that Karpelès should have realized what was going on and that his failure to do so is incredibly negligent—perhaps even criminally negligent. That doesn't change the fact that he was a victim of a group of very skilled hackers. He's the victim, not the criminal, and that's something the Japanese police should be more focused on. And they aren't. They could get to the truth if they wanted to or if they were willing to cooperate with the US government.

Federal law enforcement agencies in the United States are very interested in the Mt. Gox case, but not because they want to clear the good name of Mark Karpelès; they believe the non-Japanese hackers who looted Mt. Gox have done the same to other exchanges in US jurisdiction. In addition, they are almost certain the stolen bitcoins were laundered by a mysterious entity known as the Keyser Söze of bitcoin money laundering. They asked the Japanese authorities for the full database of Mt. Gox to further their own investigations. Their request was refused.

Unfortunately, the Japanese justice system seems to be designed less to uncover the truth than to convict someone once they have been formally prosecuted.

Japan's criminal justice system has a stunning 99 percent conviction rate (which gives an ironic twist to the lucky "1 percent"). Their success could be because their police and prosecutors are just wonderfully efficient. But six years after an effort at major reforms began, it's clear that those statistics have less to do with efficiency than forced confessions, trumped-up charges, professional judges' lack of sound judgment, and prosecutors' misconduct.

Japan's prosecutors are also notorious for junking cases that aren't pushovers, one reason why more than half of the men arrested for sexual assault or rape aren't ever prosecuted. Out of those cases, 53 percent never go to trial.

On May 21, 2009, Japan introduced a modified jury into the courtroom, in what's known as the lay-judge system. It was supposed to encourage reform and, according to Japan's Ministry of Justice, deepen the citizenry's understanding of the law.[1]

Well, the verdict is in, and the majority's opinion is this: for all practical purposes, Japanese courts operate on the principle of "presumed guilty until proven guilty."

Even if you are actually lucky enough to get a lay-judge trial and be found innocent, your innocence may last only until the prosecution appeals.

Double jeopardy? No problem in Japan, where the prosecution not only can appeal a not-guilty verdict but almost always does.

Under this new system, in cases that might result in the death penalty or other severe punishment, six lay judges are chosen from among thousands of eligible candidates of voting age to sit with three professional judges. At least one professional judge must agree with the lay judges, and a simple majority is required to determine the verdict.

According to Japan's Supreme Court, as of the end of March 2014 the lay-judge system had sentenced 6,396 people. Thirty-four of the defendants brought to trial were found not guilty, meaning precisely 99.468 percent were convicted.

Defenders of the system say this may not be as bad as it sounds. They point to a recent Supreme Court decision that told prosecutors they could no longer appeal a not-guilty ruling unless they had good reason to. "It reinforced the idea that the system is supposed to work as 'innocent until proven guilty,'" says Hiroshi Shidehara, a defense lawyer and member of the Japan Federation of Bar Association's Lay Judge System Board.

"I think it has made prosecutors less likely to challenge lay judge decisions.

"Up till now, in 90 percent of the cases where a defendant has been found not guilty, the prosecution kept appealing until they got their standard guilty verdict," says Shidehara. To the best of his knowledge, there were only four or five instances where lay judge not-guilty verdicts had been overturned successfully.

The late, great defense lawyer Makoto Endo said to me (Jake Adelstein) in 1994 over drinks that criminal trials in Japan are "absurdist farces that all end with the same last word: guilty. The prosecution is good. The judges are their deputies. The accused is a bad character who should be punished. Occasionally, one of the actors flubs his lines so badly that a not-guilty verdict results or a judge engages in a surprise act of moral conscience that changes the whole ending of the play. These mishaps are rare."

To back this up, a former Supreme Court judge has written that judges who regularly give not-guilty verdicts are liable to be sidelined or sentenced to preside over courts in remote parts of the country.

When Japan had a standard jury system, from 1928 until it was suspended in 1943, the conviction rate was around 82 percent. After the war, the system was never revived, and the new system, in which verdicts were determined by professional judges, ensured that the wheels of justice turned

with little or no resistance.

Another noted defense attorney, Hiroyuki Kawai, calls criminal cases in Japan “hostage trials.” He explains that from the time you are arrested, including the forty-eight hours you may spend in police custody, you can be held for a total of twenty-three days—and you are not guaranteed the right to see a lawyer. Your lawyer may not be present during interrogations. Your lawyer might also fail to inform you of your right to remain silent.

Meanwhile, suspects are routinely interrogated for up to eight hours a day, loosely recorded by the police. Suspects have no access to clocks. It is a breeding ground for false confessions.

“The accused is routinely denied bail and held in confinement—unless they confess,” said Kawai in an interview. “If they don’t confess, they are assumed to be susceptible of destroying evidence that would convict them, which presumes that they’re guilty in the first place. If they do confess, then later plead innocent, the judge is unlikely to believe that the confession was coerced—and they are almost sure to be found guilty.” He also notes that if a lawyer is actually unable to meet with a client, building an effective defense is next to impossible.

In reality, the prosecution has all the cards and is not even obliged to show all the evidence it possesses to the defense. According to David T. Johnson, an expert on the subject and author of *The Japanese Way of Justice: Prosecuting Crime in Japan*, there have been several cases in which prosecutors concealed critical evidence that would have exonerated the accused, but such conduct is not considered indictable.

There has been at least one case of a prosecutor tampering with evidence to convict the accused.[2] Another former prosecutor, after resigning from office, published a book in which he apologized for browbeating innocent people into making false confessions and noted that he had been educated during his training to believe that “yakuza and foreigners have no rights.”[3]

In March 2014, former boxer Iwao Hakamada was released from prison after spending forty-five years on death row. The court finally accepted DNA test results submitted by his lawyers that strongly suggested investigators had fabricated evidence. One of the original judges in the first trial admitted in 2007 that he believed Hakamada was innocent, but the judge was overruled by the two senior judges on the panel and forced to submit a guilty verdict. He resigned seven months later.

In 2003, I (Jake Adelstein) asked a functionary of the Chiba District Public Prosecutor’s Office about its refusal to accept evidence from the Belgian police that might exonerate a foreigner, Nicholas Baker, accused of

drug smuggling. The response was a look of bafflement. “Why would we want new evidence that might weaken our case, when we already have enough to convict him?” he said.

This was standard prosecution thinking: it wasn’t about justice, but about winning or saving face. Sometimes it also resulted in the guilty going free. A detective in the Criminal Investigation Department of the Osaka Prefectural Police admitted that generally the prosecution only took on a case if victory was certain. So if the accused didn’t confess, the police would drop the case, and it didn’t even go to trial. “This means that smart criminals who know the system get off.”

A good example of how unjust the system can be—and the danger of both police investigations gone bad and a process that prizes confessions over evidence—can be found in a case with some similarity to that of Mark Karpelès. This also involved an expert hacker, a cat, and some innocent people.

The Japanese police launched an investigation into a series of death threats made by a hacker, starting in the summer of 2012.[4] The lone cyber malcontent posted the threats on various Japanese websites and sent e-mails warning of violent attacks.

The investigation led to four false arrests and a major loss of face.

At the beginning of July 2012, the police made their first arrest: a nineteen-year-old student attending Meiji University in Tokyo. His prosecutor was the same as Karpelès’s, Mr. Hatsumata.

In the same month, the true hacker posted on the Osaka City website: “There will be a massacre in Osaka’s pedestrian paradise. I will run over people with a truck, stab some at random, and then kill myself.”

The post resulted in the arrest of an anime creative director who was indicted even though he continued to profess his innocence. By the end of September, the Tokyo Metropolitan Police Department (TMPD), along with the Osaka, Mie, and Fukuoka police, had arrested two more people in connection with the threats.

Seven months into the investigation, in February 2013, they finally seemed to catch the right guy.

A joint investigation task force, led by the TMPD, arrested Yusuke Katayama, a thirty-year-old IT worker, on the charge of forcible obstruction of business. He had allegedly posted a threat on the popular Internet bulletin board 2channel in August 2012, saying that a mass murder would take place at a manga convention in Tokyo.[5] This caused chaos and disrupted the event.

Katayama apparently used a virus to take over an innocent man's computer to post the message. He was suspected of committing at least three other similar crimes. Via his lawyer, he denied the charges. Although the FBI presented evidence that linked Katayama to a US server used in the crimes, with four mistaken arrests weighing on their shoulders the police proceeded with due caution.

By this stage, the search had given the lone hacker national attention as he made a fool of the country's law enforcers and challenged them to catch him with clues, taunts, false leads, and cryptic riddles.

His undoing seems to have been a final puzzle he set, which was hidden in the collar of a cat on one of Japan's islands.

At first, the police had no idea they'd been arresting innocent people. Everything changed, however, in October 2012, when e-mails claiming responsibility for the crimes were delivered first to Yoji Ochiai, a lawyer in Tokyo, and then to Tokyo Broadcasting System, as well as other media. The messages stated, "I am the real culprit," and were sent from a man using the name Oni Koroshi ("Demon Killer"). Oni Koroshi is also the name of several different brands of sake, including one cheap version much loved by the police (and reporters) and sold in packs for one hundred yen (about one dollar) in convenience stores. The e-mails contained details of how the crimes had been committed, which only the culprit could have known. Demon Killer discussed the way he had spread a Trojan horse virus (in Japan sometimes called a "remote-control virus") via online bulletin boards and then remotely controlled the host computers to post the death threats.

In his e-mail to Ochiai, he said his motive was not to enjoy seeing innocent people put in jail but "solely to trap the police and prosecutors and expose their failings to the world." He insisted that he always had intended to confess to the crimes, in due time, and save those who had been wrongly accused. He chose Ochiai because he happened to see the lawyer on TV.

The revelations in October began a cat-and-mouse game between Demon Killer and the police. The e-mails were made public, and by the eighteenth of the month, Yutaka Katagiri, chief of Japan's National Police Agency (NPA), publicly admitted that there might have been several false arrests. A day later, a new joint investigation task force was set up to get to the bottom of the crimes. By December, all four individuals who had been apprehended were cleared of all charges. On December 12, the NPA offered a reward of up to ¥3 million (approximately $36,000 at the time) for information leading to an arrest.

According to NPA sources, the cybercrime squads in each police

department had determined the IP addresses of the computers used to make the threats, but hadn't gone further to see if the machines had been affected by viruses or had malicious software installed that would make them platforms for cybercrime. According to the *Mainichi* newspaper, none of the detectives initially investigating the case even knew of the existence of remote-control viruses.

Demon Killer, aware of the police department's inability to effectively conduct a cyberinvestigation, taunted them in e-mails to the press and even direct messages saying, "Thanks for playing with me."

What made the slow investigation even more embarrassing for Japan's finest was that in two cases innocent people were pressured into making false confessions. One was the student at Meiji University, and the other a twenty-eight-year-old man in Fukuoka Prefecture.

It is clear that the police made critical mistakes at the start of the investigation, thus stretching it out over half a year, but it turns out the culprit also made some mistakes. Detectives were able to determine by November that one of the messages sent to the lawyer had gone through a US server, and they asked for the FBI's help in tracking the mail. The task force also sent some of its own people to the United States on November 12, to speed up the information-sharing process.

Sources close to the investigation say the FBI found a copy of the virus on a server in America that contained encoded information linking it to Katayama. There was also a careless mistake made in uploading the virus that allowed the route to be traced back to a computer Katayama had access to. This new evidence certainly didn't help Katayama's case. However, considering Demon Killer's ability to frame other people, it couldn't be considered conclusive.

It appears that after the media reported that investigators had been dispatched to the United States, Demon Killer began to get nervous and tried to drop out of sight. On November 13, he sent a message to the lawyer: "It's been a long time . . . I made a mistake. It looks like the game is over. It would be unpleasant to be caught so I'm going to commit suicide, by hanging myself." The accompanying photo of a witch figurine with a computer cable —in the shape of a noose, wrapped around her neck—was unsettling. The tabloids were filled with speculation as to whether the perpetrator had really killed himself.

He didn't stay dead for very long.

On January 1, 2013, he sent a traditional New Year's message to the Japanese media, encouraging them to go for a big scoop.

On January 5, he sent them new messages with a “puzzle for the New Year” to solve. The task was to find, somewhere on Enoshima Island—a popular tourist spot—a cat with a memory device in its collar that would yield clues about the man and his motives.

The police tracked down a micro SD card on a cat’s collar that same day. In the chip was the source code for the virus, with a message buried in it that said, “I was punished for a crime I wasn’t guilty of, and had to drastically rearrange my life.” One of the investigators who worked the case revealed to us later that “Katayama was already a person of interest by then because, based on previous e-mails, we were looking for someone who’d been convicted of similar crimes in the past.”

Police sources say a security camera on Enoshima captured footage of a man resembling Katayama moving toward the cat. Further examination of security cameras in the area produced footage of a motorbike that apparently belonged to Katayama. It also turned out that Katayama was a huge cat fan and made frequent visits to Tokyo’s cat cafés, where customers can play with the proprietors’ pets. This, along with other information, was used to get a warrant for his arrest.

When they searched his home, the police took away ten computers for analysis. Also, several of the computers at his workplace were found to have Tor installed—the browser designed to protect Internet privacy and make it hard for others to track any websites visited. Demon Killer had mentioned using Tor in his e-mails to the press. He may have failed to use it when he uploaded his virus.

Police suspected Katayama of using these PCs to remotely access other computers and send more than ten threats, either by e-mail or by posting them online. If Katayama was the Demon Killer, he certainly had a motive for making a fool of the police: revenge.

According to the *Mainichi*, in 2005 he had been arrested, and convicted, for posting death threats online. The threats pertained to what he perceived as an insulting illustration of a cat. To his horror, he was given a prison sentence of a year and a half in March of 2006. He wasn’t released until late 2007.

Contrast his punishment with what often happens to first-time rapists in Japan, who even if found guilty may only get a suspended sentence—as was the case recently with a Tokyo university student. It seems that only in this country can making a threat get you more prison time than actually committing a crime.

In spite of all this, no one can be absolutely certain the police finally got the right man. Ochiai, the lawyer, replied to inquiries from the press by

saying, “I can’t help feeling a little uneasy. Is this really the offender? Has there perhaps been a mistake in identification? I hope the police make doubly sure. An arrest doesn’t necessarily mean the real culprit has been caught. The police themselves have proved that point.”[6]

Words to live by.

Mark Karpelès *is* guilty of bad habits. If he had paid his huge salary into a personal account and made all his purchases from there, it would be hard to make even charges of embezzlement stick. As of late 2016, he is still waiting to be tried. Unless new charges are made, he will not be held criminally responsible for the missing 650,000 BTC, but for unrelated offenses.

When a huge business fails due to human error or a poor business plan, does someone have to go to jail? Is anyone from Lehman Brothers in jail?

In many ways, it’s hard to feel sympathy for Karpelès, and certainly the people who lost huge amounts of real and virtual money with him would agree. However, you could attribute the chain of events that resulted in the collapse of the company to his better nature: a trust in people, a desire to do the right thing and comply with the law, a refusal to give in to attempted extortion. The freezing of nearly $5 million of his US assets does seem very unjust. It would appear to be rogue agent Carl Force’s retaliation against Karpelès for not playing ball. Is that the lesson to be learned here—give in to corrupt cops or lose a fortune?

On April 17, 2015, Karpelès filed a Freedom of Information Act request with US Immigration and Customs Enforcement (ICE) for information relating to the Silk Road investigation and himself. On July 29, ICE rejected the request on the grounds that a criminal investigation was still ongoing.

On June 16, 2016, after a visit to the Tokyo District Court, where he was held in a small room and not allowed to eat, read, or do anything but wait for almost the entire day, Karpelès and his lawyer had a short meeting with the prosecution.

The next morning, when he asked a guard to post a letter to his mother before the end of the day, he was told it was impossible because the translator wasn’t there.

Months of solitary confinement awaiting trial can drive anyone crazy. This was the day Mark Karpelès lost it.

He punched and smashed his head into the wall several times, breaking a bone in a finger on his left hand. He was immediately grabbed by the guards and put in a straitjacket, like the one used when Hannibal Lecter was transferred between prisons in *The Silence of the Lambs*—arms bound, face

masked. He passed out.

He woke up in another cell, still bound, unable to move at all, and passed out again.

The inhumanity of life in a Japanese prison is a hard fact. People, particularly those not actually guilty of anything, should never be treated that way. Amnesty International has pointed this out for years. Karpelès experienced it firsthand.

On June 12, the judge granted Karpelès bail for ¥10 million, after severely criticizing the prosecution—who had tried to set bail at around ¥40 million—for dragging their feet and doing a poor job of preparing their brief. Earlier that month, at a meeting with the Cybercrime Unit, they had been told much the same: they were ignoring huge amounts of data, their case was incomplete, and their central argument of embezzlement might be wrong. In court, when asking that bail should be denied altogether, they were forced to admit that they weren't in fact ready to go to trial.

It isn't very often that a judge in Japan chews out the prosecutors.

Karpelès was finally released on bail on July 14, 2016.

Generally, government policy decisions and criminal prosecutions don't connect unless a presidential pardon is involved. However, Karpelès's complicated trial may also show that the decision by the National Diet to vote in favor of making digital currency a legal form of payment may undermine the prosecutors' argument against him.

Put simply by his lawyer: "A large part of making their case involves proving he had criminal intent. Which he didn't."

The nightmare situation Mark Karpelès found himself in—held in jail for almost a year with no trial date set, finally granted bail but unable to pay it because he had none of the bitcoins he was accused of stealing—would make anyone bang their head against a wall. Over and over again.

EPILOGUE

As the Mt. Gox fiasco fades from recent memory, banks and financial institutions are now studying and expressing interest in using bitcoin.

Coinbase, operating in California, is now one of the best-known bitcoin exchanges in the United States and handles millions of dollars' worth of transactions. In New York, financial regulators have officially recognized it as a legitimate exchange, and others have also been given the official green light.

In March 2016, the Japanese government recognized virtual currencies such as bitcoin as legal tender. The National Diet also enacted a bill to regulate operators of virtual currency exchanges in order to ensure better protection of their users. This development comes more than two years after Mt. Gox went under. At the time, bitcoin had no legal status in Japan. Mt. Gox was not required to register with the Financial Services Agency. Those in the business say the newly recognized status of their currency will obviously improve trust in it, and the companies that have replaced Mt. Gox do seem to be stronger and better run.

A huge part of the present bitcoin surge comes from China. Exchanges there offer numerous incentives, and Ponzi schemes using bitcoin help fuel the fires as well. While the Chinese stock market is slowing down, and the renminbi seems weak and unsteady, investors see an attraction in bitcoin, especially in the way it allows people to move money out of the country unnoticed.

Bitcoin exchanges also remain an attractive target for hackers.

In August 2016, Bitfinex, based in Hong Kong and the largest bitcoin exchange by US dollar volume, had $66 million worth of bitcoin looted from it. At the time of the hack, the 119,756 BTC stolen was reckoned to be worth $66 million, about a fifth of what was lost by Mt. Gox. Bitfinex survived the crisis and is still doing business.

Mt. Gox is still missing 650,000 BTC.

Where are they? The Japanese police seem to have no idea. They aren't even interested in finding them at this point; their priority is to put Mark Karpelès in jail on some charge that sticks and wash their hands of the whole mess.

Maybe Karpelès has them stashed away somewhere. Unlikely.

Shortly before this book was finished, in late July, Homeland Security Investigations, The Internal Revenue Service–Criminal Investigation

Division, and other federal agencies working together with Greek police arrested the alleged mastermind behind a $4 billion money-laundering scheme that used bitcoin transactions. The US Department of Justice identified the so-called mastermind as Russian national Alexander Vinnik, age thirty-eight. He is also known by his web moniker “WME.” The US indictment filed on July 26 alleges that Vinnik was the owner and operator of BTC-e, one of the world’s largest digital currency exchanges. The investigation revealed that BTC-e received more than $4 billion worth of bitcoin over the course of its operation. Among violations outlined in the indictment, Vinnik is accused of using BTC-e and Tradehill, another US-based exchange he owned, to process funds “obtained” from Mt. Gox between 2011 and 2014. BTC-e is believed to have processed more than 300,000 bitcoins in transactions that can be traced to the theft from Mt. Gox.

A special agent working the case stated that Mt. Gox had been the victim of a colossal hack. The authorities are hoping Vinnik will lead them to the hackers, whoever they are. We now know that over 300,000 of the hacked bitcoins have been traced but what about the other roughly 350,000 remaining unaccounted for?

Maybe, after all, angry pirate friends, a.k.a. Silk Road vendors, hacked Mt. Gox and “liberated” them.

Or maybe they’re in limbo, hidden in a box of paper wallets that Satoshi Nakamoto keeps near the Buddhist altar in his tiny ten-tatami apartment in Japan. Behind a stack of tangerines, next to the incense bowl.

Maybe Nakamoto is keeping the coins for his faithful disciple Karpelès, while the latter spends time in the limbo of the Japanese criminal justice system.

If Nakamoto is in Japan.

And if he really exists.

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the acknowledgments. If you liked the book this much, feel free to toss some bitcoins into our account. Or if you didn't like the book, you can try to hack it and prove a point!

Our bitcoin wallet is this series of numbers below. If you're really savvy, there's the QR code to boot. We have faith in the ability of people we don't know to put in bitcoin and just as much faith that people we don't know won't be able to take the money out.

The Bitcoin Jesus will (probably) bless you for your donation.



どうも有り難うございました

Nathalie and Jake

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Photo © Ayano Sato

Jake Adelstein has been an investigative journalist in Japan since 1993. Considered one of the foremost experts on organized crime in that country, he works as a writer and consultant in Japan and the United States, writing for the *Daily Beast*, the Japanese economic monthly *ZAITEN*, and other publications. He has served as a special correspondent for the *Los Angeles Times* and is the author of *Tokyo Vice: An American Reporter on the Police Beat in Japan* (Vintage), which has been translated into twelve languages, and *Operation Tropical Storm* (Kindle Single).



Nathalie Stucky is a freelance journalist in Tokyo and Europe. She was an assistant correspondent for the Japanese news agency Jiji Press in Geneva and contributed to the book *Reconstructing 3/11*. She has written for the *Daily Beast*, the *Los Angeles Times*, and several French publications.

NOTES

CHAPTER ONE

[1] Emily Spaven, “Bitcoin Enters Oxford Dictionaries Online,” CoinDesk, August 28, 2013, http://www.coindesk.com/bitcoin-enters-oxford-dictionaries-online/.

[2] The authors of this book are not computer science experts either. We have learned as we’ve worked on the story, and hopefully as you read this book, even if you have no interest in the mechanics of bitcoin, you will learn enough about it to understand how it works.

[3] United States v. Carl Mark Force IV, et al., at 9, 10 (N. D. Ca. 2015).

[4] Satoshi Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” Nakamotoinstitute.org, October 31, 2008, http://nakamotoinstitute.org/bitcoin/.

[5] In September 2015, the Kyoto police were able to arrest three men, including a college student, who used Tor to post child pornography on a members-only website that allowed purchases in bitcoin. The police used the blockchain to track them down. Law enforcement gets smarter all the time.

[6] The benefits of bitcoin mining are explained in great detail at “How Bitcoin Mining Works,” CoinDesk, last modified December 22, 2014, http://www.coindesk.com/information/how-bitcoin-mining-works/. It is highly technical, but for those who really want to know, all the nuances and “nonces” are covered.

[7] The bitcoin.org website was created with the aim of permitting individuals to register domain names without providing their own names. Even the name of the registrant of the forum where Nakamoto posted his essays is a mystery.

[8] Bitcoin true believers would probably object to comparing Satoshi Nakamoto with Dr. Victor Frankenstein for many reasons; bitcoin is not a monster and Dr. Frankenstein is a fictional creation while Satoshi Nakamoto is real. Of course, ironically we know more about the fictional Frankenstein than we do about Satoshi Nakamoto.

[9] Timothy, “Online-Only Currency BitCoin Reaches Dollar Parity,” Slashdot.org, February 10, 2011, https://news.slashdot.org/story/11/02/10/189246/online-only-currency-bitcoin-reaches-dollar-parity.

[10] Timothy, “Online-Only Currency BitCoin.”

[11] Andy Greenberg, “Crypto Currency,” *Forbes*, April 20, 2011, https://www.forbes.com/forbes/2011/0509/technology-psilocybin-bitcoins-gavin-andresen-crypto-currency.html.

[12] Leah McGrath Goodman, “The Face behind Bitcoin,” *Newsweek*, March 6, 2014, http://www.newsweek.com/2014/03/14/face-behind-bitcoin-247957.html.

[13] “Nakamoto-shi, bitcoin-kanyo o hitei” [Mr. Nakamoto denies involvement with bitcoin], *Yomiuri Shimbun*, March 10, 2014.

[14] The film is based on “In a Grove,” a 1922 story by Ryunosuke Akutagawa.

CHAPTER TWO

[1] A hosting service allows users to make their website accessible via the World Wide Web. Web-hosting companies provide space on a server owned or leased for use by their clients, as well as provide Internet connectivity, typically in a data center.

[2] The gas lines in Tokyo are automatically shut off after a major earthquake—or at least that’s what is supposed to happen.

Chapter Three

[1] This part of the Internet is intentionally hidden from search engines, uses masked IP addresses, and is accessible only with a special web browser. Sometimes it's referred to as the "dark web."

[2] James Ball, "Silk Road: The Online Drug Marketplace That Officials Seem Powerless to Stop," *Guardian*, March 22, 2013, https://www.theguardian.com/world/2013/mar/22/silk-road-online-drug-marketplace.

[3] Adrian Chen, "The Underground Website Where You Can Buy Any Drug Imaginable," Gawker.com, June 1, 2011, http://gawker.com/the-underground-website-where-you-can-buy-any-drug-imag-30818160.

[4] United States v. Ross William Ulbricht (S.D. NY 2013) [criminal complaint], https://archive.org/stream/pdfy-xl3XZpNU64Ep0zus/Ulbricht%20Criminal%20Complaint_djvu.txt.

[5] The term "white hat" in the IT world refers to an ethical computer hacker, or computer security expert, who specializes in penetration testing and other methods to ensure the security of an information system. Usually with permission, a white hat hacker will attempt to breach the security of a client's information system to find weaknesses.

[6] Andy Greenberg, "An Interview with a Digital Drug Lord: The Silk Road's Dread Pirate Roberts (Q&A)," *Forbes*, August 14, 2013, https://www.forbes.com/sites/andygreenberg/2013/08/14/an-interview-with-a-digital-drug-lord-the-silk-roads-dread-pirate-roberts-qa/#7859db457326.

Chapter Four

[1] We attempted to contact Jed McCaleb several times to ask him his version of events. While acknowledging that messages were received, he gave no response to our questions.

[2] Kim Nilsson, "The Missing MtGox Bitcoins," WizSec (blog), April 19, 2015, http://blog.wizsec.jp/2015/04/the-missing-mtgox-bitcoins.html (emphasis added).

[3] Emily Flitter, "U.S. Attorney Subpoenaed Mt Gox, Other Bitcoin Businesses: Source," *Reuters*, February 26, 2014, http://www.reuters.com/article/us-bitcoin-subpoena-idUSBREA1P18820140226.

Chapter Five

[1] Andy Greenberg, "Here's the Secret Silk Road Journal from the Laptop of Ross Ulbricht," *Wired*, January 23, 2015, https://www.wired.com/2015/01/heres-secret-silk-road-journal-laptop-ross-ulbricht/. The September 13 entry is redacted here.

[2] United States v. Carl Mark Force IV, et al. (N. D. Ca. 2015).

[3] Bay City News Service, "SF: Secret Service Agent Investigating Silk Road Admits to Theft of $820K in Bitcoin from Site," *Contra Costa Times*, September 1, 2015.

[4] Adrianne Jeffries, "US Government Seized $5 Million from Bitcoin Behemoth Mt. Gox," *Verge*, August 23, 2013, https://www.theverge.com/2013/8/23/4651926/us-government-seized-5-million-from-bitcoin-behemoth-mt-gox.

[5] Andy Greenberg, "DEA Agent Charged with Acting as a Paid Mole for Silk Road," *Wired*, March 30, 2015, https://www.wired.com/2015/03/dea-agent-charged-acting-paid-mole-silk-road/; Nick Bilton, "Silicon Valley Murder Mystery: How Drugs and Paranoia Doomed Silk Road," *Vanity Fair*, April 26, 2017, http://www.vanityfair.com/news/2017/04/silk-road-ross-ulbricht-drugs-murder; Bay City News Service, "SF: Secret Service Agent Investigating Silk Road Admits to Theft of $820K in Bitcoin from Site," *Contra Costa Times*, September 1, 2015.

[6] Olivier Janssens, Twitter post, https://twitter.com/olivierjanss/status/555889044618346498, January

15, 2015.

Chapter Six

[1] Andy Greenberg, “Here’s the Secret Silk Road Journal from the Laptop of Ross Ulbricht,” Wired,

[2] Jonathan Soble, “Mark Karpeles, Chief of Bankrupt Bitcoin Exchange, Is Arrested in Tokyo,” *New York Times*, August 1, 2015, https://www.nytimes.com/2015/08/02/business/dealbook/mark-karpeles-mt-gox-bitcoin-arrested.html?_r=0.

[3] This figure is according to a press release given to the Japanese media. The actual amount in October 2013, when the embezzlement allegedly occurred, would have been upward of $3.2 million.

[4] “Karupuresu hikoku saitaiho e” [Police to rearrest defendant Karpelès], *Yomiuri Shimbun*, October 28, 2015.

Chapter Seven

[1] If you want to understand more about Japan’s modified jury system, the Japanese Ministry of Justice website has a good overview (http://www.moj.go.jp/ENGLISH/).

[2] Dustin Dye, “Osaka Prosecutor Pleads Guilty to Tampering with Evidence,” *Foreign Policy Blogs*, March 19, 2011, https://foreignpolicyblogs.com/2011/03/19/osaka-prosecutor-pleads-guilty-of-tampering-with-evidence/.

[3] “Public Prosecutors Were Taught That Yakuza & Foreigners Have No Rights,” *Japan Probe*, May 26, 2011, http://www.japanprobe.com/2011/05/26/public-prosecutors-were-taught-that-yakuza-foreigners-have-no-rights/ (page discontinued).

[4] Jake Adelstein and Nathalie-Kyoko Stucky, “How Japan’s Cyberterrorist Lost Game of Cat and Mouse,” *Daily Beast*, February 25, 2013.

[5] 2channel was the inspiration for America’s much-reviled discussion board, 4chan. The creator of 2channel is allegedly now the de facto owner of 4chan. Read more at Jake Adelstein, “Will 4Chan’s Shady New Owner Weaponize It?,” *Daily Beast*, September 26, 2015, http://www.thedailybeast.com/will-4chans-shady-new-owner-weaponize-it.

[6] Adelstein and Stucky, “How Japan’s Cyberterrorist Lost.”